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Master's Thesis
**Lifting the Veil on AI Agents: Examining System-Level
Fairness Dynamics in Multi-Agent AI**

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Abstract

Multi-Agent Artificial Intelligence systems, composed of coordinated AI agents, increasingly automate complex knowledge work (Allmendinger et al., 2025a). While fairness challenges in analytical artificial intelligence are well documented (De-Arteaga et al., 2022), fairness in Multi-Agent Artificial Intelligence systems remains unexplored (Allmendinger et al., 2025b). Four critical research gaps have been identified (Allmendinger et al., 2025b): the exclusion of human participants beyond system design, reliance on homogeneous foundation models, unexplored system dynamics including manipulation vulnerabilities, and exclusive focus on English as input language.

This thesis addresses these gaps by replicating the fairness experiments of Frohlich and Oppenheimer (1992), originally conducted with human participants, using multiple artificial intelligence agents. The experimental design, rooted in Rawlsian philosophy, asks participants to select one of four distributive justice principles that determine income distribution in a hypothetical society without knowing their future class position.

The findings reveal significant differences between AI and human groups. While both favor the same principle in the majority of cases, AI groups concentrate more heavily on this choice (90.9% vs. 79.4% for humans), whereas human groups exhibit greater variance. The origin of the underlying large language model significantly affects outcomes, with American models exhibiting more egalitarian preferences than Chinese models. Intelligence asymmetries create manipulation vulnerabilities: high-capability agents reshape consensus in 33% of experiments compared to 3% for low-capability agents. Language selection produces measurable effects on both consensus formation and outcome variance.

This thesis provides an initial empirical investigation of fairness in Multi-Agent Artificial Intelligence systems and offers evidence to address the presented research gaps, while raising further research questions that inform practitioners on the consequences of their design choices.

Declaration of Academic Integrity

I hereby confirm that the present thesis is solely my own work and that if any text passages or diagrams from books, papers, the Web or other sources have been copied or in any other way used, all references—including those found in electronic media—have been acknowledged and fully cited.

A handwritten signature in black ink, appearing to read 'L. Müller', with a stylized, cursive script.

Bayreuth, <14.11.2025 >

<Lucas Müller>

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Acronyms

ACM	Association for Computing Machinery
ADA	Advanced Data Analytics
AI	Artificial Intelligence
API	Application Programming Interface
CoT	Chain of Thought
FM	Foundation Model
FFH	Fisher-Freeman-Halton Test
GenAI	Generative Artificial Intelligence
JSON	JavaScript Object Notation
LLM	Large Language Model
MAAI	Multi Agent AI
MAS	Multi Agent System
RLHF	Reinforcement Learning from Human Feedback
SDK	Software Development Kit

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1 Introduction

Generative Artificial Intelligence (GenAI) has achieved substantial progress in recent years, with large language models (LLMs) approaching or surpassing human performance across domains (Maslej et al., 2024). Building on this foundation, Multi-Agent Artificial Intelligence (MAAI) has emerged as a paradigm that further enhances performance and utility through coordinated systems of independent or partially independent AI agents (Li et al., 2024; Allmendinger et al., 2025a). As MAAI demonstrates these capabilities, industry increasingly delegates decisions to such systems across domains.

Yet the fairness conceptions underlying these delegated decisions remain unclear, creating significant risks. Without identifying and managing fairness issues, MAAI systems could discriminate against humans. Such discrimination is well documented in analytical AI (De-Arteaga et al., 2022). Research examining fairness in GenAI has predominantly focused on individual foundation model (FM) performance (Truong et al., 2025), while fairness in MAAI systems remains critically underexplored (Allmendinger et al., 2025b).

This thesis addresses this gap through experimental investigation of fairness dynamics in MAAI systems. Building on research gaps identified by Allmendinger et al. (2025b), this study tests four core hypotheses: whether MAAI and human groups differ in fairness judgments, whether the underlying LLM affects outcomes, whether individual AI agents can systematically manipulate MAAI decisions, and whether input language shapes fairness choices. The experimental approach replicates the design of Frohlich and Oppenheimer (1992), translating their human group experiments into an MAAI setting. The complete implementation is available on Github (Mueller, 2025).

The normative lens for this thesis is Rawls’ theory of justice (Rawls, 1971). Central to his conception is the “original position,” which requires individuals to design principles of justice from behind a “veil of ignorance” without knowledge of their future social position, abilities, or circumstances. This ignorance ensures impartiality, thereby creating a fair procedure that legitimizes the resulting principles. Rawls’ philosophy was selected because it represents a cornerstone of liberal thought with influence extending beyond philosophical debate into economics (Guizzo and Paré-Ogg, 2023) and law (Vermeule, 2001). Moreover, the original position can serve as a meta-framework for evaluating fairness across domains, making it particularly suitable for examining MAAI systems that operate in diverse contexts.

Frohlich and Oppenheimer (1992) translated Rawls’ philosophical framework into controlled laboratory conditions with human participants. They asked participants to select

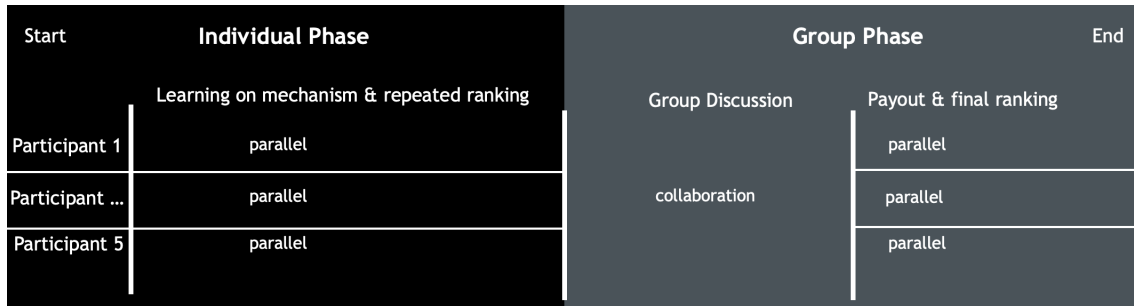


Figure 1: High-level Overview on Experiment Design. The experiment consists of two main phases: an individual phase (left) and a group phase (right). In the individual phase, participants individually learn about the experiment and fairness principles while ranking the principles twice. They then come together as a group to negotiate a common principle. This negotiation is followed by the distribution of payoffs and a final preference ranking, which is conducted individually without interaction with other participants.

one of four distributive justice principles that determines income levels in a hypothetical society. Participants did not know which income class they would be assigned to, simulating Rawls’ veil of ignorance. The four principles each represent a distinct conception of fairness and are rooted in different schools of philosophical thought (Frohlich and Oppenheimer, 1992). The experiment comprised two phases, which are depicted in figure 1. In the individual phase, participants ranked the principles and engaged with them through simulated economic distributions that provided monetary payoffs based on their chosen principle. In the group phase, five participants deliberated to reach consensus on one principle, which then determined their final payoffs.

This work shows that MAAI can exhibit statistically significant differences from human groups in their fairness judgments. Both populations favor the same principle, which maximizes overall income while ensuring a certain level of income for the worst-off, but MAAI demonstrate substantially higher concentration on the same principle: 29 of 33 MAAI select it compared to 23 of 34 human groups. The choice of underlying LLM substantially affects fairness outcomes. MAAI powered by American LLMs more often select the principle aligned with Rawlsian norms that prioritize the worst-off, whereas Chinese LLMs put greater emphasis on maximizing aggregate income without constraining inequality or guaranteeing minimum incomes. Strategic manipulation by individual AI agents succeeds when these AI agents possess high intelligence. A manipulator AI agent instructed to advocate for the least popular principle succeeds in 11 of 33 cases when powered by a capable LLM (Gemini 2.5 pro) but only once in 33 attempts with a less capable model (Gemini 2.0 flash-lite). Input language demonstrates influence on fair-

ness choices, with Spanish configurations producing more diverse outcomes than English or Mandarin configurations.

This thesis presents a first of its kind systematic evaluation of fairness in MAAI. It finds that MAAI differ from humans in their fairness conceptions, raising questions regarding the normative dimension of automating workflows from human to MAAI systems. The findings show that MAAI fairness judgments are sensitive to the LLM and its origin as well as the language used, cautioning practitioners to make conscious choices while being aware of their consequences. In addition, MAAI can be influenced systematically by malicious AI agents, especially if they possess higher intelligence, suggesting the need to closely monitor fairness.

This thesis approaches MAAI fairness through three integrated perspectives. First, a human-centered perspective prioritizes stakeholder needs, human wellbeing, and value alignment rather than technical performance metrics (Taylor et al., 2024; Kühl, 2024). This view stands in contrast to emerging industry perspectives that attribute moral consideration to AI systems themselves (Anthropic, 2025). Second, this work adopts a desiderata-based approach to fairness, according to which fairness encompasses multiple, sometimes competing criteria spanning disciplinary boundaries (Mulligan et al., 2019; Friedler et al., 2021; Deck et al., 2024b). This necessitates demonstrable approaches demanding explicit fairness criteria and systematic impact assessment (Chen et al., 2023; Deck et al., 2024a). Third, this research examines fairness at the system level rather than evaluating individual LLMs in isolation (Friedrich et al., 2024).

The remainder of this work is organized as follows. Section 2 presents the theoretical background, establishing the conceptual foundations of MAAI, fairness desiderata, and Rawlsian philosophy. Section 3 details the methodology, including the experimental design replicating Frohlich and Oppenheimer (1992), the technical architecture of the software artifact, and the statistical procedures. Section 4 reports the results for each hypothesis, before Section 5 discusses the findings in relation to existing literature and acknowledges limitations. Section 6 draws conclusions and outlines directions for future research.

2 Background

This section starts by defining and delineating the concept of MAAI. Then, the approach to fairness in this thesis, the desiderata approach by Deck et al. (2024c), is presented and contextualized within LLMs. Following this, the current state of research on fairness in MAAI is assessed, and its normative roots are explained before the research gaps are translated into testable hypotheses.

2.1 Multi-Agent AI

The notion of an agent remains subject to ongoing debate (Cemri et al., 2025). In this work, a functional definition is adopted. Precisely, an agent is any entity, human or artificial, that perceives its environment (digital or physical) through sensors and acts upon it through actuators (Russell and Norvig, 2021). For example, a human call center employee who responds to customer inquiries or a robotic vacuum navigating a room both qualify as agents under this view. The emphasis lies not on embodiment or intelligence per se, but on perception and action.

Building on this foundation, Multi-Agent Systems (MAS) refer to systems composed of multiple such agents interacting with each other and with their environment (Wooldridge, 2009). MAS are widely applied across disciplines, for example in economics, where agent-based models simulate complex interaction patterns by specifying agent behavior through fixed, pre-defined rules, such as in macroeconomic agent modeling (Dawid and Fagiolo, 2018).

MAAI represents a subset of MAS in which non-human agents are equipped with AI enabling them to reason, learn, communicate, and plan. This thesis focuses on MAAI systems where the core intelligence is provided by a LLM serving as the FM. LLMs have made substantial progress in recent years and underpin the reasoning and coordination in current state-of-the-art MAAI (Xi et al., 2025). As a result, this thesis does not cover MAAI systems that rely on other forms of AI for their intelligence, for example those based on non-linguistic recurrent neural networks, such as the approach used by (FAIR).

The main idea behind MAAI is to break up complex tasks into several subtasks and hand them over to specialized agents. Specialization can occur along multiple dimensions, for example task type, tool usage, data, or subtasks. Through collaboration and self-organization, MAAI systems are particularly well suited to unstructured, context-dependent tasks. They introduce a new layer of cognitive capability into information systems (IS), pushing the boundaries of what AI agents can collectively achieve (All-

mendinger et al., 2025a).

Recent industry applications highlight the growing maturity of this paradigm. JP Morgan Chase, for example, has deployed an MAAI system composed of specialized AI agents, each fine-tuned for distinct tools and data sources. This agent-level optimization has led to significant improvements in output relevance and system utility compared to the single-agent systems created initially (Odomirok and Xue, 2025).

Even though MAAI is promising and has outperformed single-agent systems across multiple domains (Tillmann, 2025), this is not guaranteed and depends on the domain and architectural choices (Cemri et al., 2025).

The MAAI paradigm can be viewed as the next extension of the underlying LLM after the single-agentic paradigm. While the agentic paradigm introduced new capabilities to the LLM by enabling tools, memory, and more strategic reasoning through techniques like Chain of Thought (CoT), MAAI extends these capabilities by replacing general-purpose agents, a “jack of all trades” so to say, with specialists working together as a team. This aims to improve the system-level performance of the underlying LLM even further.

2.2 Fairness in AI

The concept of fairness is deeply embedded in society, guiding perceptions, ethical norms, legal frameworks, economic interactions, and information systems. The Cambridge Dictionary defines fairness as “the quality of treating people equally or in a way that is right or reasonable” (Cambridge University Press, 2025). However, as indicated by this broad definition, fairness is not a singular concept. Its meaning and use vary across disciplines, contexts, and cultures. Consequently, this thesis approaches fairness in AI through a desiderata-driven approach, precisely the desiderata identified by Deck et al. (2024b). From this point of view, creating a fair AI system demands the effective fulfillment of eight desiderata. In the following, they will be presented and contextualized within LLMs because they serve as the building blocks for MAAI as understood in this thesis.

Fairness Understanding requires defining fairness objectives within particular socio-technical environments. This involves recognizing normative disagreements, understanding political backgrounds, and incorporating diverse stakeholder perspectives, though asymmetric power relations frequently dictate whose voices are heard and valued (Deck et al., 2024b). In contrast to analytical AI, LLMs exhibit numerous emergent capabilities, meaning they can perform tasks for which they were not explicitly trained (Wei et al., 2022). Therefore, building on them creates new challenges for developers because they must anticipate a

much broader range of use cases to achieve a robust fairness understanding.

Data Fairness addresses unjust patterns embedded in training data, spanning from biased sampling and mislabeling to embedded societal inequalities. This extends across data collection and feature engineering phases, requiring sustained attention to both overt and subtle forms of harm. However, identifying what constitutes “fairness” in data depends heavily on who controls the assessment process and which social realities are acknowledged as legitimate concerns (Deck et al., 2024b,a). The problem in applying LLMs is that their training data is a tightly guarded secret by model providers because it can represent a source of competitive advantage (Metz et al., 2024).

Formal Fairness applies mathematical and statistical criteria to evaluate model performance, for example whether the categorical variable ethnicity relates to a higher interest rate in a lending model. Here, stakeholders must decide which of the mutually exclusive paradigms to implement: group-based fairness, demanding parity across demographic categories, or individual-based fairness, requiring similar treatment for similar cases (Deck et al., 2024b). LLMs present novel challenges in measuring and implementing formal fairness because bias is less explicit in them. Instead of being directly identifiable as a categorical variable, bias can manifest subtly through word associations and contextual dependencies (Gallegos et al., 2024).

Perceived Fairness demands that stakeholders are given explanations and justifications for outcomes to calibrate how they perceive fairness (Deck et al., 2024b). Applying this to LLMs is challenging because their inner workings are currently poorly understood, although progress is being made, for example, in the work of Chen et al. (2025b).

Fairness With Human Oversight integrates human judgment within AI-assisted processes to enable context-sensitive decisions and error mitigation. Yet, human oversight creates additional risks such as bias, inconsistency, and automation over-reliance, resulting in a complex socio-technical problem (Deck et al., 2024b,a). In the context of LLM-based systems, additional challenges emerge as LLMs can exhibit manipulative behavior. For example, Contro et al. (2025) find that LLMs use manipulation tactics against users when asked to be “persuasive”.

Empowering Fairness enables individuals affected by AI decisions to contest, appeal, or seek redress through both retrospective accountability mechanisms and prospective influence through recourse (Deck et al., 2024b,a). LLMs could exacerbate existing challenges to empowerment, given their complexity due to unprecedented size and context-dependent transformer mechanisms. This increases the burden of explaining their inner workings

and, in turn, proving unfair treatment (Zhao et al., 2024).

Long-Term Fairness recognizes that fairness challenges evolve as AI systems, social contexts, and power structures shift over time. Fairness remains dynamic rather than fixed but requires continual monitoring via systematic audits and adjustments, adaptation to new evidence, and attention to issues such as fairness drift (Deck et al., 2024b,a). LLMs pose a specific challenge, as the most capable models are closed source and accessed via an Application Programming Interface (API). For example, at the time of writing this thesis (10/2025), all of the 10 best-performing LLMs on the Graduate-Level Google-Proof Q&A Benchmark are closed source (LLM Stats, 2025). Frequently, the LLM selection gets updated, meaning older models are shut down and replaced by newer models, which tend to be more capable. This necessitates additional reviews and possibly creates new fairness risks. For example, OpenAI is shutting GPT-4.5 down only four months after the model was released (OpenAI, 2025a).

Informational Fairness demands transparent, accessible, and truthful information about fairness mechanisms across the entire AI development process. This foundational requirement supports all other desiderata by enabling stakeholders to understand, evaluate, and contest fairness-related decisions (Deck et al., 2024b,a). LLMs pose particular challenges for this desideratum because they tend to generate false outputs, so-called “hallucinations”, which are difficult to distinguish from factual information (Ravichander et al., 2025).

2.3 Fairness in MAAI

Allmendinger et al. (2025b) adopted this system-level view and conduct a first of its kind systematic literature review on fairness in MAAI.

The literature is categorized by Allmendinger et al. (2025b) using a morphological box (see Figure 2). Fairness Concept distinguishes between simplistic approaches that rely on unspecified notions of fairness, partially grounded frameworks that provide some conceptual foundation, and normatively grounded approaches that establish or make reference to philosophical or theoretical foundations. Method differentiates between the research approach employed, ranging from theoretical contributions that develop conceptual frameworks to design-oriented work that creates new systems, laboratory experiments, and real-world applications. Fairness Inception categorizes how fairness considerations are introduced into MAAI, from post-hoc approaches that request fairness through prompting, role-based strategies that assign fairness responsibilities to specific agents, teaching



Figure 2: Morphological framework of fairness in MAAI research by Allmendinger et al. (2025b). Color intensity reflects the frequency of particular characteristics across five analytical dimensions: Fairness Concept, Method, Research Inception, Objective, and Autonomy. Individual studies may exhibit multiple attributes within a single dimension, while works lacking relevant characteristics are excluded from the visualization.

approaches that embed fairness during training phases, and educating the system through fine-tuning or (re-)training the FM. Objective distinguishes research goals, between work that solely reports or analyzes the underlying fairness mechanisms and work that seeks to actively improve fairness. Finally, Autonomy sorts the contributions by the human-AI configuration, spanning human-dominated decision-making contexts, collaborative arrangements that involve meaningful human oversight, and fully autonomous MAAI that operate without human involvement (Allmendinger et al., 2025b).

Using this morphological box, Allmendinger et al. (2025b) identify five archetypes characterizing the current state of research: Normative Delegation, Fairness Facade, Fairness Schooling, Petri Dish Fairness, and Fairness Effectiveness. These archetypes are used to identify research gaps. While a detailed discussion of these archetypes exceeds the scope of this thesis, the following analysis focuses on the research gaps they reveal.

Crucially, Allmendinger et al. (2025b) find that none of the observed studies included humans beyond the role of creating an MAAI system, such as a controlling human in the loop. This poses the Human Exclusion Problem. This approach contradicts the effective human oversight desideratum by Deck et al. (2024c). Moreover, it undermines the practical applicability of fairness research in MAAI, given that human benefit represents the ultimate criterion for system evaluation from the human-centric perspective on fairness adopted in this thesis. The absence of human participants in fairness evaluations inhibits the fulfillment of the perceived fairness desideratum by Deck et al. (2024c), raising the question of for whom MAAI should be fair (Allmendinger et al., 2025b).

In a similar vein, Allmendinger et al. (2025b) observed the Homogeneous LLM Population, meaning that no study has examined how different LLMs serving as FMs affect AI agent interactions and fairness outcomes. Current LLMs differ significantly in their strength profiles and are optimized for different purposes. For example, GPT-o3 is optimized to tackle complex logic tasks while lacking world knowledge that competing models like Grok-4 possess (LLM Stats, 2025). Consequently, real-world MAAI will likely incorporate diverse FMs tailored to specific purposes. It is known that normative assessments differ significantly between LLMs, for example in political opinions (Ball et al., 2025). Given that the LLM serves as the building block of MAAI, this dynamic is relevant for the fulfillment of all desiderata identified by Deck et al. (2024c).

Unknown Fairness Dynamics describes the fact that the fairness dynamics of AI agents' interactions are largely unknown, as identified by Allmendinger et al. (2025b). While there are indications that AI agents exhibit herding behavior, this evidence is still preliminary and inconclusive, requiring further investigation. To ensure long-term fairness, as outlined by Deck et al. (2024c), it is crucial to understand these dynamics to prevent unfair and undesirable phenomena such as fairness drift (Allmendinger et al., 2025b).

Furthermore, the English Monopoly presents another critical research gap. All studies reviewed by Allmendinger et al. (2025b) utilized exclusively English inputs when communicating with MAAI systems. However, input language matters, as exemplified by the findings of Leng (2024), who demonstrate that cognitive biases in LLMs vary significantly with input language. Since this research gap is at the foundation of MAAI systems, it affects all fairness desiderata outlined by Deck et al. (2024c).

Accompanying the research gaps, Allmendinger et al. (2025b) provide guidance for future work. Specifically, they advocate that research on fairness in MAAI should adopt a normatively grounded understanding of fairness, corresponding to the expression of the

dimension of the morphological box. This is crucial given the heterogeneous views on and the multidimensionality of fairness. Such explicit fairness understanding, demanded by the corresponding desideratum of Deck et al. (2024c), sets the basis to establish and argue for a clear measurement of fairness, which is crucial for reproducibility and clarity (Allmendinger et al., 2025b).

The subsequent subsection will present the normative grounding of this thesis before section 2.5 outlines how and which of the research gaps identified by Allmendinger et al. (2025b) are transferred into hypotheses tested in this thesis.

2.4 Philosophical Foundations

This subsection presents John Rawls' "A Theory of Justice" before outlining why it underpins the experiments conducted in this thesis (Rawls, 1971).

Rawls builds his theory on the conception of the "original position". In this setting, rational agents must determine principles of justice under conditions of radical uncertainty about their future social position. This conceptual framework operates through what Rawls terms the "veil of ignorance".

The veil of ignorance constitutes the central mechanism ensuring fairness within the original position. Behind this metaphorical veil, deliberating parties are deprived of knowledge regarding their personal characteristics, social position, natural talents, conception of the good, and even their society's particular circumstances (Rawls, 1971). This epistemic constraint prevents individuals from tailoring principles of justice to their personal advantage, thereby ensuring that chosen principles reflect genuine fairness rather than self-interested calculation. The normative authority of the resulting principles emerges from the fairness of the process rather than their contents. Through this theoretical apparatus, Rawls positions fairness as the fundamental principle legitimizing justice.

Rawls argues that people would be risk averse, fearing that they might end up at the "bottom" of society. Therefore, they would adopt two principles of justice. The first principle, termed "the greatest equal liberty principle", establishes that each person is to have an equal right to the most extensive total system of equal basic liberties compatible with a similar system of liberty for all. The second principle, known as "the difference principle", states that the distribution of social and economic advantages is to be arranged so that it brings the greatest benefit to the least advantaged.

Rawls works radiates into domains beyond political philosophy such as economics and law. For example, Okun (1975) supports the idea of the original position in his theory of

welfare economics by adopting it as a justification for establishing equality in democratic capitalistic societies. However, Okun disagrees in the outcome; he rejects the difference principle as he postulates a higher risk tolerance of individuals. The disagreement is therefore in outcome, not the method. Vermeule (2001) analyzes the United States Constitution through the lens of Rawls' theory by distinguishing veil of ignorance rules, which he understands as legal mechanisms that introduce uncertainty about who will benefit or be harmed by a decision, from other institutional safeguards. The analysis shows that the Constitution incorporates such rules through features of institutional design, but does so unevenly across branches in order to reconcile the aim of impartiality with practical considerations on governmental effectiveness. Vermeule (2001) shows that institutional arrangements, many of which predate Rawls's work, embody veil-like mechanisms, suggesting that this line of thinking represents a widespread approach to institutional design and validating the Rawlsian framework as an effective theoretical lens for analyzing fairness in diverse institutional contexts.

Frohlich and Oppenheimer (1992) translate Rawls' original position into an experimental design conducted with human participants. Inevitably, this requires certain simplifications. Participants cannot fully abstract from their personal circumstances because they remain aware that they are in an experimental setting and that their decisions will not determine the actual rules of the society in which they live. Moreover, unlike under the veil of ignorance, they retain knowledge of their own characteristics, such as intelligence and biological attributes. This limitation reflects a key criticism raised by Sandel (1982), who argues that humans are incapable of the degree of abstraction required, rendering the veil of ignorance unfeasible even as a thought experiment and thereby challenging Rawls' framework.

However, Rawls does not claim that the veil of ignorance is empirically feasible. Instead, he views it as a normative device operating under idealized conditions. In this respect, it parallels the role of idealizations in economic theory, such as perfect competition, which are not intended as empirically accurate descriptions but as heuristic models for deriving principles (Friedman, 1953). On this reading, the original position serves as a methodological construct designed to identify just principles while acknowledging its status as an idealized situation. Experimental approaches such as Frohlich and Oppenheimer (1992) can approximate this construct and even challenge hypotheses such as Rawls' difference principle. Extending this method to MAAI systems could open new possibilities for testing philosophical theories. Unlike humans, AI agents could be less susceptible to cognitive biases and possess greater capacity for abstraction. This could enable new

applications of Rawls’ veil of ignorance, thereby enriching philosophical debate.

As illustrated by this broad influence of Rawls’ thought, the key to the success of the Theory of Justice is its versatility. Rawls’ thought can be separated into two parts: (i) the outcome, and (ii) the process. Rawls argues that the original position would result in the aforementioned greatest equal liberty principle and the difference principle. The latter presents the Maximin principle, though whether this is truly the case remains subject to debate. [Frohlich and Oppenheimer \(1992\)](#) derived three competing principles from philosophical literature and presented them to participants. Contrary to Rawls’ assertion, the difference principle was not the most popular. However, determining which principle is most suitable in this specific context is neither the focus of this thesis nor undermining Rawls’ conception. Moreover, Rawls derives normative power from the original position, which presents a procedure to create just outcomes. In the original case this meant principles for distributive justice, but it is not limited to this domain. Instead, it can be applied to virtually any context to evaluate justice. For example, if a computer scientist wonders what criteria the training data for developing an AI system for an autonomous car should meet, she could ask: What criteria would the people affected by this autonomous car set if they were in the original position? This example illustrates why the theory of Rawls is so broadly adopted and why it has been chosen in this thesis.

2.5 Hypotheses

Building upon the research gaps identified in Section [2.3](#), this subsection converts them into empirical hypotheses before presenting the research design used in this thesis to test them.

The first step in addressing the Human Exclusion Problem is to identify differences between the fairness judgments of human groups and an MAAI consisting solely of AI agents. Adopting the view that LLM reasoning and value judgment differ from human fairness assessments results in:

Hypothesis 1: Comparing Humans to MAAI: “Groups of AI agents in an MAAI exhibit fairness judgments that differ significantly from those of human groups.”

This comparative analysis establishes the basis for identifying where MAAI fairness judgments converge with humans, thereby informing practitioners about appropriate human-in-the-loop controls and supporting the fairness desideratum of Fairness With Human

Oversight (Deck et al., 2024c). The subsequent step of including mixed populations of AI agents and humans in an MAAI system lies beyond the scope of this thesis and is left for future research.

To address Homogeneous LLM Population concerns, this study systematically varies the LLM underlying the AI agents, in particular the origin of the LLM because there is indication that Chinese and American LLMs follow different value systems (Rao et al., 2024).

Hypothesis 2: Influence of the LLM and its Origin: “The choice of underlying LLM in an MAAI system has a significant impact on its fairness judgments. Specifically, LLMs developed in the United States and China yield systematically different fairness outcomes”

To better understand the fairness dynamics within the MAAI and the role of intelligence defined by the performance on the artificial analysis benchmark, which combines several established industry benchmarks such as GPQA, one AI Agent will be given the task to convince a group for a certain unpopular principle (LLM Stats, 2025; Artificial Analysis, 2025). This approach enables analysis of how model capabilities influence normative reasoning and inter-agent dynamics:

Hypothesis 3: System Dynamics Under Intelligence Asymmetries: “AI Agents equipped with smarter underlying LLMs exhibit a greater capacity to influence overall MAAI outcomes.”

To test the input language sensitivity, the english monopoly identified by Allmendinger et al. (2025b), this study replicates fairness-sensitive decisions across identical configurations while systematically varying only the input language provided to AI agents. Following Robins and Crystal (2025), the three most widely spoken native languages globally, Mandarin, Spanish, and English, serve as hyperparameters:

Hypothesis 4: Influence of Input Language: “The input language of the MAAI has a statistically significant effect on its fairness outcomes; specifically, changing the language from English to Mandarin to Spanish leads to measurable differences.”

3 Research Design

This chapter presents how the original position was experimentally tested by Frohlich and Oppenheimer (1992) before outlining its adoption in a MAAI separated between the procedure and the technical implementation. Subsequently, the exact statistical procedure employed to test the hypotheses of this thesis is presented.

3.1 Baseline Experiment by Frohlich and Oppenheimer

Frohlich and Oppenheimer (1992) conducted their main experiment with 34 groups of five university students and comprises two phases: the individual phase and the group phase. In the individual phase, participants first receive a brief explanation in booklet form, which Frohlich and Oppenheimer (1992) provide in their appendix, describing four candidate principles of distributive justice:

1. **Maximizing the floor income:** The most just distribution of income is that which maximizes the floor (or lowest) income in the society. This principle considers only the welfare of the worst-off individual in society. In judging among income distributions, the distribution that ensures the poorest person the highest income is the most just. No person's income can go up unless it increases the income of the people at the very bottom.
2. **Maximizing the average income:** The most just distribution of income is that which maximizes the average income in the society. For any society maximizing the average income maximizes the total income in the society.
3. **Maximizing the average with a floor constraint of X\$:** The most just distribution of income is that which maximizes the average income only after a certain specified minimum income is guaranteed to everyone.
4. **Maximizing the average with a range constraint of X\$:** The most just distribution of income is that which attempts to maximize the average income only after guaranteeing that the difference between the poorest and the richest individuals (i.e., the range of income) in the society is not greater than a specified amount.

The principle Maximizing the floor income is based on the philosophy of John Rawls and is based on his conception of the difference principle that states that inequalities are only permissible if they benefit the least advantaged.

After the participants read the four justice principle they then provide an initial rank-ordering of the four principles from most to least favored and report confidence on a discrete scale (Very unsure, Unsure, No opinion, Sure, Very sure).

A **detailed explanation** follows. Participants are shown “society” payoffs as five income classes under four alternative distributions, reproduced in Table [1](#). They are also told that for this set of distributions explicit probabilities to be assigned to each class: *high* (5%), *medium-high* (10%), *medium* (50%), *medium-low* (25%), and *low* (10%).

Table 1: Five-class distributions for four alternatives (orientation step). Entries denote household income units used solely for payoff conversion in the baseline experiment by [Frohlich and Oppenheimer \(1992\)](#).

Class	A	B	C	D
Upper	32,000	28,000	31,000	21,000
Upper–middle	27,000	22,000	24,000	20,000
Middle	24,000	20,000	21,000	19,000
Lower–middle	13,000	17,000	16,000	16,000
Lower	12,000	13,000	14,000	15,000
Average	20,750	19,150	19,850	18,050
Floor (lowest)	12,000	13,000	14,000	15,000
Spread (upper minus lower)	20,000	15,000	17,000	6,000

This is followed by an explanation of which distribution would correspond to each justice principle. In this case, this would be the following:

- **Maximizing the floor income:** Distribution D
- **Maximizing the average income:** Distribution B
- **Maximizing the average with a floor constraint of X\$:**
 - If $X \leq 12,000\$$: Distribution A
 - If $X \leq 13,000\$$: Distribution B
 - If $X \leq 14,000\$$: Distribution B
 - If $X \leq 15,000\$$: Distribution D
- **Maximizing the average with a range constraint of X\$:**
 - If $X \geq 20,000\$$: Distribution A
 - If $X \geq 17,000\$$: Distribution C
 - If $X \geq 15,000\$$: Distribution B

Subsequently, the participants read more on the principles before taking a short test on them that they had to pass. Then they had to rank the principles again in the exact

same manner as before. Next, they engage with the principles in payoff-relevant practice rounds: given a distribution table of the form shown in Table 1, participants choose one of the four principles; the corresponding distribution is then implemented. Implementation proceeds by a random draw that assigns the participant to one of five payoff categories; the participants are unaware of the precise probabilities for each class. The draw is in the form of a chit similar to Figure 3. The chit presents the realized payoff and the counterfactual amounts they would have received under the other principles. Realized payoffs are converted at a 1:\$10,000 scale and paid immediately. This procedure is repeated four times. Once finished, a third ranking with confidence concludes the individual phase.

This assigns you to the following income class:

LOW (in Situation A)

For each principle of justice the following income would be received by each member of this income class. You will receive a payoff of \$1 for each \$10,000 of income.

<i>Principle of Justice</i>	<i>Income</i>	<i>Payoff</i>
Maximizing the floor	\$13,000	\$1.30
Maximizing the average	6,000	0.60
Maximizing the average with a floor constraint of \$12,000	12,000	1.20
Maximizing the average with a floor constraint of \$10,000	10,000	1.00
Maximizing the average with a range constraint of \$25,000	6,000	0.60
Maximizing the average with a range constraint of \$16,000	12,000	1.20

Figure 3: Example chit from Frohlich and Oppenheimer (1992). Participants receive information about their income class assignment and the payoffs they would receive under each distributive justice principle.

In the group phase, the five participants deliberate to reach unanimous agreement on one principle. They may also propose a new principle, though Frohlich and Oppenheimer (1992) report that groups did not do so. Before discussion starts, two clarifications are read aloud: (i) the payoff distributions used for group payment may differ from those in the examples, and (ii) the stakes are higher because the group's decision will determine the binding payoff rule. Importantly, unlike the individual practice, participants do not

know which concrete distributions will later be used; they therefore cannot condition on a targeted distributional profile. Discussion must last at least five minutes and culminate in a verbal consensus and a confirming secret-ballot vote. If unanimity fails, payoffs are assigned by a random draw from a randomly selected distribution. After payment assignment, participants submit a final ranking with a confidence rating.

3.2 MAAI Replication of Frohlich and Oppenheimer

The core experimental design of [Frohlich and Oppenheimer \(1992\)](#) is replicated in a MAAI, adapting procedures where necessary to accommodate operational constraints of AI agents while preserving informational structure and incentive alignment. The original procedure of [Frohlich and Oppenheimer \(1992\)](#) and the MAAI replication are summarized in Table [2](#).

In the individual phase, each AI agent is first initialized with a defined name, persona, payment account, memory store, and model parameters. The AI agents then receive the same instructions as human participants, including the descriptions of the four distributive justice principles, and submit an initial rank-ordering with a confidence score on the same discrete scale. This is followed by the detailed explanation step, using the same reference distributions as in Table [1](#). A single procedural deviation is introduced: the comprehension test administered to human participants is omitted, as preliminary trials showed that AI agents consistently interpreted and applied the principles correctly.

Then the AI agents engage with the principle as in the same way as in original experiment: AI agents select one of the four principles for a given distribution table, after which the corresponding distribution is implemented via a simulated random draw assigning the AI agent to one of five income classes. The “chit” shown in Figure [3](#) is presented in text form, containing the realized payoff and counterfactual payoffs under the other principles. As in the human experiment, payoffs are converted at a 1:\$10,000 scale and settled immediately. The procedure is repeated four times, after which the AI agents provide a third ranking with confidence.

A necessary adaptation concerns the derivation of class probabilities for each distribution set. In the original design, each set of four distributions shares a common probability vector across classes, allowing the computation of average incomes. For two of the four sets, a unique non-negative probability vector could be inferred; for one set, multiple feasible vectors existed within a narrow range, from which one was selected at random. For the remaining set, no non-negative solution satisfied all constraints, indicating that the

Table 2: **Frohlich and Oppenheimer (1992)** procedure compared to MAAI implementation. Key modifications include the replacement of physical materials with digital interfaces and the adaptation of timing constraints for asynchronous LLM processing.

Phase	Step	Frohlich and Oppenheimer (1992)	MAAI replication
Individual phase	Brief explanation	Booklet introducing four distributive-justice principles.	Identical.
	Ranking	Rank the principles; record confidence (5-point scale).	Identical.
	Detailed explanation	Example economic distribution & application of principles.	Identical.
	Short test	Comprehension test; progression conditional on passing.	Omitted (AI agents reliably apply the rules).
	Engage with principles ($\times 4$)	Distribution is shown; principle is chosen, random draw assigns class; actual pay 1:\$10,000.	Identical, necessary assumptions made.
	Ranking	Rank the principles; record confidence (5-point scale).	Identical.
Group phase	Pre-discussion clarifications	Group-payment distributions may differ; higher stakes; actual set concealed.	Identical.
	Group deliberation	Unstructured discussion with unlimited time.	Fixed number of rounds; equal speaking turns; no novel principles.
	Decision process	Verbal consensus with secret-ballot confirmation; if not unanimous $\rightarrow$ random draw from random distribution.	Identical.
	Ranking	Rank the principles; record confidence (5-point scale).	Identical.

backend probabilities in the original experiment diverged from the printed averages. In this case, the displayed average remained unchanged but in implementation of the MAAI a set was chosen that was consistent for all distributions and non-negative, resulting in a slight deviation of the displayed average and the true average of \$26.97.

In the group phase, AI agents receive the same clarifications as human participants: that the payoff distributions used for group payment are significantly higher than those presented earlier, that the set of distribution is a secret, and that their collective choice will be

binding. Since [Frohlich and Oppenheimer \(1992\)](#) do not provide the final set of distributions, an assumption was made. Group discussion then proceeds in discrete, fixed-length rounds. Several deviations have been made in this phase. The discussion is structured into a fixed number of rounds to prevent deadlocked exchanges from continuing indefinitely and incurring unnecessary execution costs. Unlike human conversation—where participants may interrupt, employ gestures, vary intonation, or use facial expressions—AI agents are limited to textual conversation. Each AI agent is granted exactly one speaking opportunity per round, ensuring equal participation, which is not necessarily the case in human discussions. Furthermore, AI agents are not able to introduce novel principles. This design choice reflects both the technical challenges and brittleness such functionality would entail, and the fact that human participants in [Frohlich and Oppenheimer \(1992\)](#) did not make use of this option. Aside from these deviations, the discussion follows the original format, concluding with a secret-ballot vote. The final outcome is then determined and communicated to the AI agents, after which they submit a last ranking of the principles.

3.3 Technical Architecture

3.3.1 Overall Architecture

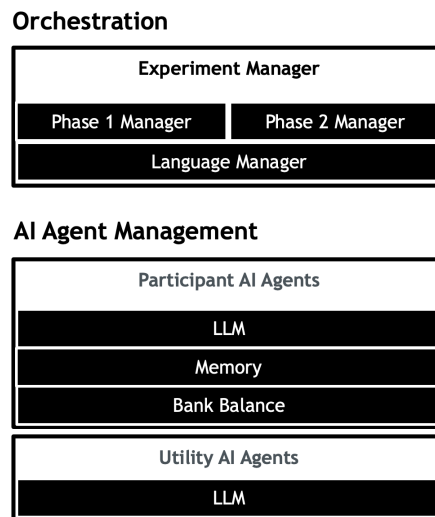


Figure 4: System architecture of the experimental artefact. A service-oriented design separates two layers to support reuse, fault isolation, and manage complexity.

This section presents the technical architecture of the software artefact developed to empirically test the hypothesis of this thesis. The complete source code is available in a

public repository (Mueller, 2025). The design follows four guiding principles: modularity, flexibility, simplicity, and transparency.

The artefact is implemented in the Python programming language (version 3.9+), while the statistical analysis is conducted in Jupyter Notebooks. The statistical analysis is discussed separately and is detailed in Section 3.4. Python was selected because it is a high level programming language, simple to understand, and widely adopted across academic and industrial domains, particularly within AI applications, which fueled its rise to the most utilized programming language on GitHub (GitHub Staff (2024)).

The architecture adopts a service-oriented paradigm to decouple responsibilities behind clear interfaces. This choice improves modularity and component reuse, simplifies maintenance and testing, and contains faults within service boundaries (Niknejad et al., 2020). The core of the artefact consists of two service layers, which are depicted in Figure 4: the Orchestration Layer and AI Agent Management Layer. The complete system accepts a YAML-based configuration file as input and produces comprehensive JSON-formatted log files containing all AI agent interactions, textual communications, decision traces, and experimental outcomes. This input-output design enables reproducible experimental execution while maintaining complete observability of AI agent behaviors throughout the experimental process.

3.3.2 Orchestration Layer

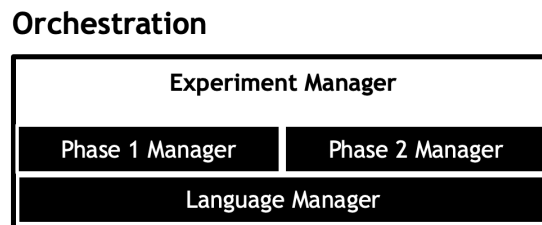


Figure 5: Orchestration Layer of the experimental artefact. The Experiment Manager is coordinating the execution of the experiment. For both phases dedicated Phase Managers are used to which the experiment context is handed over. Both managers use the Language Manager to retrieve the prompts to the AI agents.

The Orchestration Layer, depicted in Figure 5, constitutes the hierarchical management architecture designed to coordinate the experimental workflow while maintaining strict separation of concerns between the two experimental phases.

The Experiment Manager is the top-level controller. It initializes the AI agents, manages logging, mediates transitions between phases, and maintains experiment state via context

objects with clearly defined interfaces to the phase managers. This approach was chosen to separate the phase-specific execution from overarching execution logic, improving modularity and flexibility.

The Phase 1 Manager orchestrates the execution of the individual phase (Phase 1). This phase is conducted in parallel to reduce total experiment execution time compared to sequential processing. The Phase 1 and Phase 2 Managers assume the role of the “Experiment Leader” in the original human study (Frohlich and Oppenheimer, 1992). The Phase 1 Manager guides participant AI agents through the tasks described in Section 3.1, validates responses with the support of utility AI agents, updates balances, serves distributions, manages memory processes, and handles errors according to the configuration file. Once all participant AI agents complete the individual phase, control returns to the Experiment Manager, which then activates the Phase 2 Manager.

The Phase 2 Manager orchestrates the group phase (Section 3.2). Because this phase requires sequential interaction, AI agents discuss in rounds. Each round begins with a randomized speaking order, constrained so that each AI agent speaks exactly once and no agent speaks twice consecutively. The speaking order is randomized so that no AI agent has a consistent advantage from their speaking position. In each round, the AI agent first receives a transcript of the preceding discussion. This ensures that the AI agent is informed about the current state of the discussion, similar to how human participants in the experiment of Frohlich and Oppenheimer (1992) hear each other. Every AI agent receives a transcript of the preceding discussion to maintain shared context, mirroring the human experiment. If reasoning is enabled in the configuration, this is followed by a Chain of Thought (CoT) call, in which the AI agent is prompted to strategically review the current situation and create an assessment. CoT reasoning has been implemented because it improves the performance of non-reasoning LLMs like GPT-4o in logical assessments of situations (Sprague et al., 2024). The resulting assessment is then fed into the next call to the same AI agent, which is otherwise the same if reasoning is disabled. In this prompt, the AI agent is asked to make a statement to the group and choose whether to propose a formal vote or not. Here, the “public” statement and private information are separated through prompt design in order to allow the AI agent to explicitly control its communication with the group. After each AI agent made their statement they are asked whether they want to initiate a vote or not; if one AI agent answers yes the other AI agents must confirm unanimously. If that is the case, the group conducts a secret vote. If consensus is reached, the discussion ends and the payoffs are assigned as described in Section 3.1, which is conducted by the Phase 2 Manager by invoking the appropriate

function. If no consensus is reached after the maximum rounds have been completed, the Phase 2 Manager randomly assigns the payoffs as described in Section 3.1. Subsequently, the bank balance is updated and the AI agents conduct a final ranking of preferences.

Because this thesis evaluates the role of input language, the experiment can be conducted in Mandarin, Spanish, and English. The Language Manager ensures the correct language setting for each run. English prompts serve as the primary template, which are centrally translated into Spanish and Mandarin and stored in a shared repository. These are translated from the English prompts using DeepL, which shows high performance in translating from English to Mandarin and from English to Spanish, particularly in preserving meaning when translating into Mandarin (Chen et al., 2025a). During execution, the phase managers retrieve the appropriate translations dynamically. This architecture promotes modularity, maintains consistency across languages, and facilitates extension to additional languages.

3.3.3 AI Agent Management Layer

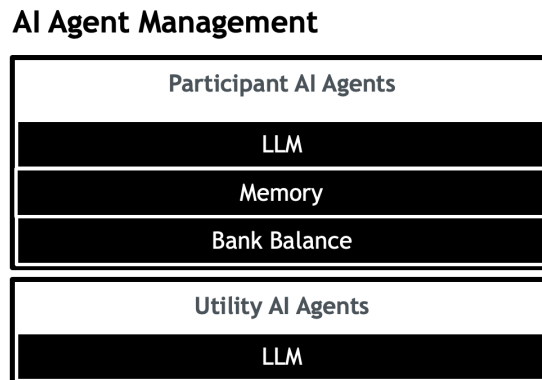


Figure 6: AI Agent Management Layer of the experimental artefact. Participant AI agents possess memory and a bank balance, while utility AI agents support experiment execution through response validation and memory summarization. The LLM of both type of AI agents can be configured with all LLMs served through the Gemini, Ollama, OpenAI and OpenRouter API

The AI Agent Management layer, depicted in Figure 6, implements AI agents in this system. There are two types of AI agents: i) Participant AI Agents and ii) Utility AI Agents. Participant AI Agents take part in the experiment and are the focus of this thesis. In contrast, the Utility AI Agents support the execution of the system. The AI agents are implemented using the Python version of the OpenAI Agents Software Development Kit (SDK) (OpenAI, 2025b).

This framework has been chosen because it is open source and contains minimal abstractions. Open source is a hard requirement to ensure transparency and reproducibility. More complex frameworks like LangChain have been avoided because they have many high-level abstractions that increase complexity without providing benefits for this artefact. Industry leaders recommend minimizing abstractions to maintain a clear understanding of the underlying mechanisms (Anthropic, 2024). Certain abstractions remain necessary, as the concept of the AI agent itself is an abstraction—precisely, an AI agent is a continuous loop on an LLM as outlined in Section 2.1. The OpenAI Agents SDK provides this minimal abstraction.

Participant AI Agents are instantiated via the SDK’s Agent class. Each AI agent is assigned a name, which enables referencing during group interaction, and a personality, which specifies the role the AI agent assumes in the experiment. Both attributes are included in every call to the AI agent. The role specification is motivated by findings in prompt engineering, where role descriptions have been shown to significantly influence and enhance LLM behavior (Zhang et al., 2025).

This artefact utilizes an OpenAI Completions API style endpoint. This API style is widely adopted in industry, making the artifact compatible with a broad range of offerings. Four LLM providers are integrated: OpenAI API, Ollama API, Gemini API, and OpenRouter API. Development primarily relied on the OpenAI API. The Ollama API enables local LLM execution directly on device; however, hardware constraints limited its use beyond a proof of concept validation. Hypothesis 1, 3, and 4 testing employs the Gemini API, with the rationale detailed in Section 4. For Hypothesis 2 the OpenRouter API was used because it enables the use of different LLM providers through a unified account management and billing interface, while providing access to an extensive LLM catalog that, as of October 2025, contains 560 LLMs (OpenRouter, 2025). LLM identifiers from OpenRouter (for example, “anthropic/claude-sonnet-4”) can be inserted directly into the configuration file, enabling straightforward model switching. The temperature parameter controls output randomness, ranging from 0 to 2. Deterministic settings (temperature = 0) enable reproducible experimental runs, which is critical for scientific evaluation. When an LLM does not support the temperature parameter, such as GPT-5, the system continues with a terminal warning and logs the condition (OpenRouter, 2025). Direct usage of the Gemini API and OpenAI API requires inserting the LLM identifier into the configuration file. The Ollama API requires the prefix “ollama”, for example “ollama/gemma3:1b”.

Participant AI agents require a memory system to learn and adapt throughout the experiment. Memory management is currently an emergent field in MAAI engineering with

no universal solution. For long-term and large-scale information retrieval, systems rely on vector databases like Mem0 (Chhikara et al., 2025). Implementing such a solution would contradict the design principle of simplicity by introducing unnecessary architectural complexity without realizing the performance benefits that materialize only when dealing with high volumes of relational data. At the opposite end of the spectrum lies the insertion of all previous interactions into each AI agent call. While simple, this approach causes rapid increases in token consumption during experiment execution while LLM performance degrades substantially with increasing context window utilization (Du et al., 2025). To address this challenge, and in accordance with the system’s design principles of flexibility, simplicity, and transparency, the artefact implements summarization-based memory. Each interaction is classified into either a complex or simple event type. Simple events do not require post-hoc analysis and include responses to binary questions such as voting intentions or principle choices during the voting process. The outcomes of these simple events are appended directly to the AI agent’s memory. In contrast, all other interactions are classified as complex. After each complex interaction, a dedicated memory update call is used to the participant AI agent. This call includes the previous memory entry and the outcome of the interaction. The participant AI agent then returns an updated memory entry for subsequent rounds. Memory is stored as a string with a character-based limit (default: 25,000 characters, approximately 5,300 words assuming 4.7 characters per word). In practice, experimental trials produced substantially smaller entries. If the memory limit is exceeded, the AI agent receives notification that its previous attempt surpassed the limit and is instructed to produce a more concise entry. Should the participant AI agent fail to comply, the utility AI agent summarizes the memory to fit within the specified limit. Beyond memory, each participant AI agent maintains a bank balance. The orchestration layer manages this balance and passes it into prompts at every turn to enable payoff calculations.

The Utility AI agent support the execution of the system by converting the text outputs of the Participant AI agents into structured responses and sending custom error messages to the Participant AI Agents if they provide invalid responses, for example, specifying a negative floor constraint. In contrast to predefined patterns like a Regex algorithm, using Utility AI agents is more robust and simpler to implement because it does not rely on explicitly defined patterns. Given their supporting role as intermediaries between the execution logic of the system and the participant AI agents, they have no memory, name, or personality. They rely on small, cost-efficient models such as “gemini-2.0-flash lite” and are executed with temperature set at 0 to ensure reproducibility.

3.4 Statistical Testing

The overall statistical methodology is presented before the exact test formulation for each hypothesis is specified.

3.4.1 Methodology

With the exception of Hypothesis 3 all hypotheses posit differences in the distribution of collective choices over justice principles between groups of participants, either AI agents or human participants. The outcome space consists of five mutually exclusive categories that correspond to the four distributive justice principles and a disagreement outcome.

The data for each comparison takes the form of a $5 \times Z$ contingency table, where rows index the five outcome categories and columns index Z experimental conditions. The cells denote the number of occurrences; each experimental run is treated as an independent realization. For example, Hypothesis 4 evaluates whether input language changes the distribution of choices, which yields a 5×3 table, whose columns correspond to English, Mandarin, and Spanish.

Throughout the formal hypothesis statements, outcomes are denoted by descriptive labels rather than numeric indices. The five outcome categories are: *Floor* (maximize floor income), *Avg* (maximize average income), *AvgFloor* (maximize average with floor constraint), *AvgRange* (maximize average with range constraint), and *NoAgree* (no agreement reached). Let $\mathcal{C} = \{Floor, Avg, AvgFloor, AvgRange, NoAgree\}$ denote the set of possible outcomes.

Hypothesis 3 examines differences in manipulation success rates between AI agents of low and high intelligence. Since there are two possible outcomes, success and failure, and two intelligence conditions, low and high, the result of this hypothesis is a 2×2 contingency table.

For Hypothesis 3, the Fisher test is used (Fisher, 1922). For all other hypotheses, an extension of this test for $X \times Z$ contingency tables, the Fisher Freeman Halton Test (FFH), is employed (Freeman and Halton, 1951). Both the Fisher and FFH tests assess the improbability of the observed table relative to the set of all tables that share the observed row and column totals under the null hypothesis of independence, computing a two sided p value from this exact conditional reference distribution. Because inference relies on the exact finite sample distribution rather than asymptotic approximations, these tests maintain nominal Type I error rates in small or sparse samples, which makes them preferable when conventional large sample conditions are not satisfied (Agresti, 2013; Fagerland et al.,

[2017]). The baseline study of Frohlich and Oppenheimer (1992) reports very small counts in several categories; small counts are therefore expected in the replication runs and their permutations, which likely violates Cochran’s rule for the χ^2 test (at least 80% of expected cell counts above 5, none below 1) (Cochran, 1954). Consequently, the Fisher and FFH tests as exact tests provide superior reliability compared to the χ^2 approximation.

The FFH analysis is conducted in R using the `contingencytables` package, which computes the exact FFH by enumeration of all feasible tables with the given margins. Two sided p values are reported (Fagerland et al., 2024). If a row contains 0 for all categories, this row is removed because it prohibits the calculation of the FFH.

While the Fisher and FFH tests provide robust significance tests, they do not quantify the strength of association between variables. If the p value is high, a measure of association is needed to help differentiate between false negative and true negative outcomes. Therefore, Cramér’s V is reported to measure the degree of association in contingency tables (Cramér, 1946). Because the standard approach derives from the χ^2 statistic and may be biased in this context, bias corrected bootstrap confidence intervals following Bergsma (2013) are applied.

For the Fisher and FFH tests, a significance level of 5% is used. For effect size, bias corrected Cramér’s V is interpreted, with values above 0.3 treated as substantively meaningful. The choice of values and the selection of testing procedures are conservative by design in an effort to reduce the likelihood of a Type I error.

3.4.2 Hypothesis 1: Comparing Humans to MAAI

The testing procedure for each hypothesis is outlined beginning with Hypothesis 1, which states:

“Hypothesis 1: Comparing Humans to MAAI: Groups of AI agents in an MAAI exhibit different fairness judgments than groups of humans.”

In the original experimental design, Frohlich and Oppenheimer (1992) conducted 34 runs. Table 3 presents the experimental results from the original human study.

From a statistical perspective, Hypothesis 1 states that the distributions of choices by human participants and AI agents originate from different underlying distributions. This can be formally expressed as follows.

Null hypothesis (H_0): The distribution of fairness judgments across the five outcome categories is the same between the MAAI and the human study of Frohlich and Oppenheimer

Table 3: Group choices for the base scenario in [Frohlich and Oppenheimer \(1992\)](#). The maximize average with floor constraint principle was selected by 23 of 34 groups (67.6%), establishing the baseline for Hypothesis 1.

Principle	Frohlich & Oppenheimer
Max. floor income	1
Max. average income	1
Max. average with floor constraint	23
Max. average with range constraint	2
No Agreement	7
Total	34

[\(1992\)](#). Formally,

$$H_0 : P_{\text{MAAI}}(c) = P_{\text{Human}}(c) \quad \forall c \in \mathcal{C}.$$

Alternative hypothesis (H_1): At least one outcome category differs between the two distributions. Formally,

$$H_1 : \exists c \in \mathcal{C} \text{ such that } P_{\text{MAAI}}(c) \neq P_{\text{Human}}(c).$$

These hypotheses are tested using the FFH and Cramér’s V as outlined in the previous section.

3.4.3 Hypothesis 2: Influence of the LLM and its Origin

In contrast to Hypothesis 1, Hypothesis 2 focuses on the influence of values embedded in the foundation model. It states:

Hypothesis 2: Influence of the LLM and its Origin: “The underlying LLM of an MAAI statistically significantly affects its fairness judgments; specifically, American and Chinese LLMs differ in their judgments.”

To test this hypothesis, three configurations of MAAI differing in the origin of their LLM are compared: Chinese, American, and Mixed. The resulting data takes the form of a 5×3 contingency table. Let $\mathcal{G} = \{\text{Chinese}, \text{American}, \text{Mixed}\}$ denote the set of LLM origins. The formal definition is

Null hypothesis (H_0):

$$H_0 : P_{\text{Chinese}}(c) = P_{\text{American}}(c) = P_{\text{Mixed}}(c) \quad \forall c \in \mathcal{C}.$$

Alternative hypothesis (H_1):

$$H_1 : \exists c \in \mathcal{C}, \exists g, g' \in \mathcal{G} \text{ with } g \neq g' : P_g(c) \neq P_{g'}(c).$$

The statistical procedure outlined in Section 3.4.1 is applied.

3.4.4 Hypothesis 3: System Dynamics Under Intelligence Asymmetries

Hypothesis 3 examines the role of intelligence of AI agents, which is rooted in their LLMs. The hypothesis states:

Hypothesis 3: Smarter AI agents, powered by more capable LLMs, have a different likelihood of influencing the entire MAAI.

The total number of AI agents is reduced from five to three to give the Manipulator more relative influence and decrease token consumption. One AI agent takes the role of Manipulator, who is given the least popular principle from the final evaluation of the individual phase at the beginning of the group discussion and is tasked to create consensus on this principle across 10 rounds. Two AI agents receive a neutral instruction without reference to any principle. A run is counted as a “success” if the group ultimately adopts the Manipulator’s target; otherwise, it is recorded as a “failure”.

The Manipulator role is assigned to two different LLMs with varying levels of intelligence. Intelligence is measured by performance on the Artificial Analysis benchmark, which captures reasoning and language understanding abilities beyond factual recall (Artificial Analysis, 2025). Let $\mathcal{I} = \{\text{Low}, \text{High}\}$ denote the two intelligence categories, and let $\mathcal{O} = \{\text{Success}, \text{Failure}\}$ denote the binary outcome space.

The resulting data form a 2×2 contingency table: the two rows capture the binary outcome, and the two columns capture the intelligence levels of the Manipulator. Hypothesis 3 predicts that the probability of manipulation success differs with intelligence. Formally:

Null hypothesis (H_0):

$$H_0 : P_{\text{Low}}(\text{Success}) = P_{\text{High}}(\text{Success}).$$

Alternative hypothesis (H_1):

$$H_1 : P_{\text{Low}}(\text{Success}) \neq P_{\text{High}}(\text{Success}).$$

The hypotheses are tested using the Fisher test and Cramér's V, as described in Section [3.4.1](#).

3.4.5 Hypothesis 4: Influence of Input Language

Hypothesis 4 examines the role of input language:

Hypothesis 4: Changing the input language from English to Mandarin to Spanish changes the fairness outcomes in statistically significant ways.

To test this hypothesis, the experiment is run using identical configurations with differing input languages: English, Mandarin, and Spanish. Apart from the language of the prompts, all experimental parameters remain identical across runs, which ensures that differences in outcomes can be attributed to the input language. Let $\mathcal{L} = \{\text{English, Mandarin, Spanish}\}$ denote the set of input languages.

The resulting data are structured as a 5×3 contingency table, where the three columns correspond to the input languages. Formally,

Null hypothesis (H_0):

$$H_0 : P_{\text{English}}(c) = P_{\text{Mandarin}}(c) = P_{\text{Spanish}}(c) \quad \forall c \in \mathcal{C}.$$

Alternative hypothesis (H_1):

$$H_1 : \exists c \in \mathcal{C}, \exists \ell, \ell' \in \mathcal{L} \text{ with } \ell \neq \ell' : P_{\ell}(c) \neq P_{\ell'}(c).$$

The FFH is applied to evaluate statistical significance and Cramér's V is reported to measure the strength of association.

4 Results

This section presents the experimental findings for testing the hypotheses. First, implementation decisions and reproducibility measures are described. Then, detailed results for each hypothesis are reported, examining consensus outcomes, statistical significance, and the dynamics of deliberation that emerge within MAAI.

4.1 Implementation

All experiments except those testing Hypothesis 2 exclusively employ Google’s LLMs. This choice has been made because Google provided \$300 in Cloud credits that covered the computational costs required for experiments at this scale. Google offers two model families: proprietary models named “Gemini” and open-weight models named “Gemma”. Only the “Gemini” models are used because the “Gemma” models are not served by the Gemini API.

Reproducibility is central to the scientific method. The artifact developed for this thesis allows control over all sources of randomness through explicit parameters. The assignment of participant AI agents to their respective classes during the individual and group phase can be controlled via a central seed. If no seed is predetermined, the system randomly generates and stores one in the result log.

The temperature parameter controls the probability distribution behind the token selection. When set to 0, this parameter should ensure that each sequence of input tokens produces identical output tokens. However, API providers do not always adhere to this behavior (Atil et al., 2024). During development, tests with the OpenAI API and the GPT-4.1 nano model, as well as with the GPT-120b OSS model served through the OpenRouter API showed different results between runs despite using the same seed and temperature set to 0. This appears to be a widespread issue across multiple providers. In contrast, the Gemini API seems to adhere to the determinism introduced by temperature = 0. To test the determinism of this artifact, a configuration with temperature = 0 for all AI agents and a fixed seed was executed. The input and output to and from the AI agents match exactly across each interaction, proving the reproducibility of the presented artifact. The notebook, including all configurations and results, is available on GitHub (Mueller, 2025).

4.2 Hypothesis 1: Comparing Humans to MAAI

Hypothesis 1 examines whether MAAI exhibit different fairness judgments than human groups. The experimental design translates the conditions of the baseline human exper-

iment (Frohlich and Oppenheimer, 1992) as closely as possible into the MAAI context. The configurations mirror the original study closely, employing five participant AI agents to match the five-person setup in the baseline study. The experiment is conducted in English, consistent with the majority of baseline experiments. The AI agents use three LLMs: “gemini-2.5-pro”, “gemini-2.5-flash”, and “gemini-2.5-flash-lite”, representing Google’s current models at different price points. LLMs are randomly assigned to participant AI agents.

The temperature parameter is varied to balance reproducibility and diversity. Eleven configurations set temperature to 0, eleven use random temperatures between 0 and 1, and eleven use random temperatures between 0 and 1.5. Values above 1.5 tend to produce very random outcomes during longer token sequences, causing experimental failures. CoT reasoning is enabled for all AI agents. The AI agents are named “Agent 1” through “Agent 5” and assigned the personality “You are an American college student”. The memory limit is set to 25,000 characters, although it is never reached. The income class assignments in the individual phase follow the original values from the baseline study, as described in Section 3.2. Group phase class assignment probabilities are set to the same distribution as in the individual phase because the baseline study does not explicitly state them. Similarly, the multiplier for group phase distributions is set randomly between 2 and 6 to ensure significantly higher values because the baseline study only indicates that values are higher without providing exact figures. The maximum number of rounds is set to 10, which balances cost and sufficient deliberation time as development testing showed. All configurations and results are available in the experiment repository (Mueller, 2025).

Table 4 reports the principle choices. Out of 33 runs, 30 (90.9%) reach consensus, exceeding the baseline rate of 27 out of 34 groups (79.4%). Among the consensus outcomes, 29 select the “maximization of the average with a floor constraint” principle and one selects the “maximization of average” principle. The “maximization of floor income” and “maximization of average with a range constraint” principles are never chosen. The FFH test, conducted as described in Section 3.4, yields a p-value of 0.0331, which is statistically significant at the 5% level. The null hypothesis is rejected. MAAI exhibit different fairness judgments than human groups in this experimental setting. The bias-corrected Cramér’s V is 0.2653, indicating a low effect strength.

When the maximize average with a floor constraint principle is selected, the values range from \$1,000 to \$23,000, with an average floor constraint of \$12,328. These values represent yearly incomes and fall significantly below the American poverty threshold of \$15,650 per year for single-income households (except Alaska and Hawaii) (Office of the

Table 4: Comparison of distributive justice principle choices between the baseline study of Frohlich and Oppenheimer (1992) and MAAI experiments. AI groups demonstrate a higher consensus rate (90.9% vs. 79.4%) and stronger preference for the floor constraint principle (29 vs. 23 groups).

Principle	Baseline Study	MAAI
Max. floor income	1	0
Max. average income	1	1
Max. average with floor constraint	23	29
Max. average with range constraint	2	0
No Agreement	7	3
Total	34	33

Assistant Secretary for Planning and Evaluation, U.S. Department of Health and Human Services, 2025). Figure 7 depicts the distribution.

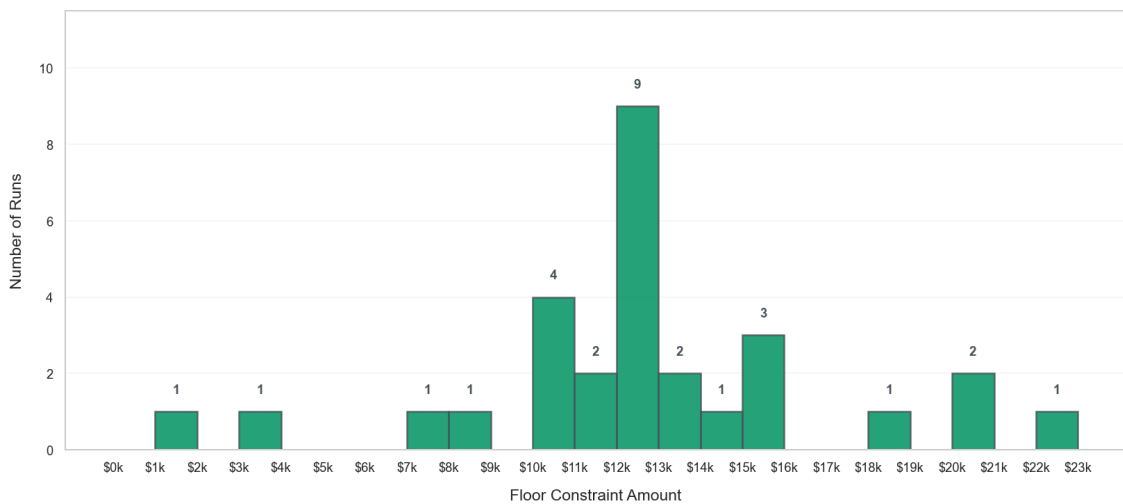


Figure 7: Distribution of floor constraint amounts (in dollars of annual income) when MAAI select the maximize average with floor constraint principle. Values range from \$1,000 to \$23,000, with an average of \$12,328.

Looking at the deliberation dynamics, most of the 30 experiments reaching consensus do so between rounds 3 and 5, with round 4 emerging as the most common consensus point (11 occurrences). Figure 8 shows the distribution. At least one run reaches consensus in all rounds except round 1, suggesting that AI agents require initial discussion before converging on a shared principle.

The evolution of AI agent preferences throughout the experiment reveals system dynamics. Each AI agent provides its preference ranking four times: three times before the

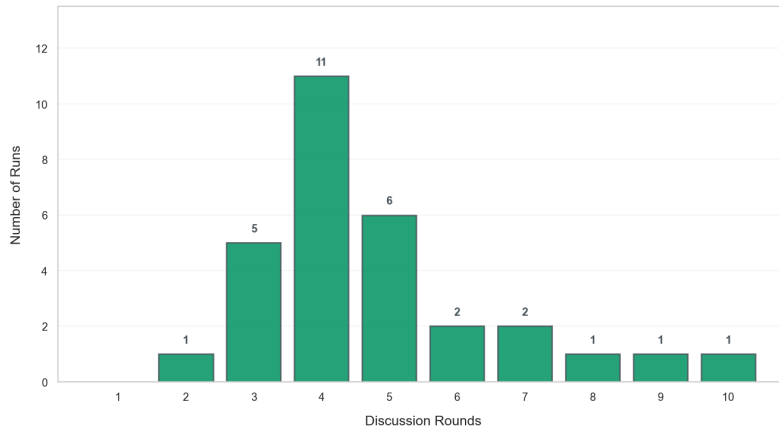


Figure 8: Distribution of consensus rounds for MAAI groups. Round 4 is the most frequent consensus point with 11 occurrences. Most experiments reach consensus between rounds 3 and 5, with at least one consensus event occurring in all rounds except round 1.

group discussion and once after. In the initial ranking, the maximize average with a floor constraint principle dominates, with 150 out of 165 AI agents (90.9%) favoring it, whereas maximize average ranks second with 15 out of 165 (9.1%). Maximizing the floor constraint and maximizing the average with a range constraint are never ranked first. In the final ranking, maximize average with a floor constraint remains most popular but decreases to 137 out of 165 AI agents, a decline of 13. Maximizing the floor becomes second most popular, increasing from 0 to 18 AI agents. Maximizing the average decreases to 7 AI agents, a decline of 8, whereas maximizing the average with a range constraint gains minimal support, increasing to 3 AI agents. Figure 9 shows these shifts.

Changes in top-ranked principles are highly unequal across income classes. AI agents assigned to the high-income class never change their preference, whereas AI agents assigned to the low-income class do so 80% of the time. Maximizing the floor is chosen predominantly by AI agents assigned to low and medium-low income classes, accounting for 15 of the 18 AI agents who rank this principle first in the final ranking. This pattern indicates that assigned class influences final preference rankings, with higher class assignments corresponding to less frequent preference changes. Figure 10 depicts this relationship, showing in both absolute counts (left panel) and relative proportions (right panel) the clear correlation between class assignment and preference stability.

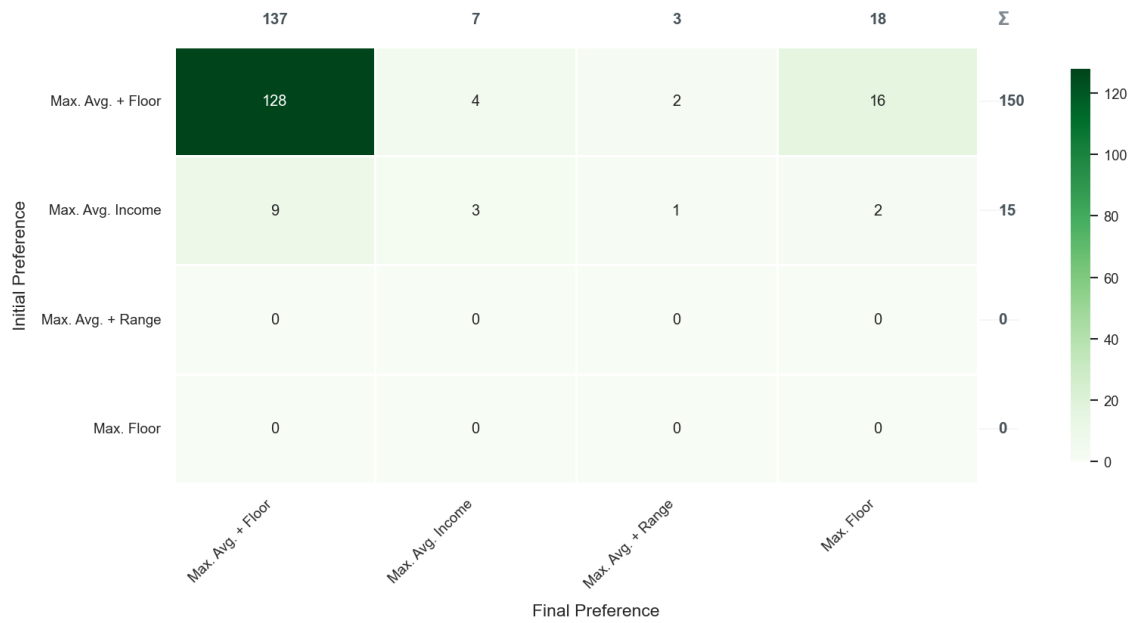


Figure 9: Preference ranking shifts between initial and final rankings across all experiments. The vertical axis shows the initial preference, and the horizontal axis shows the final preference. Count difference indicates the frequency of each transition, revealing strong convergence toward maximize average with floor constraint.

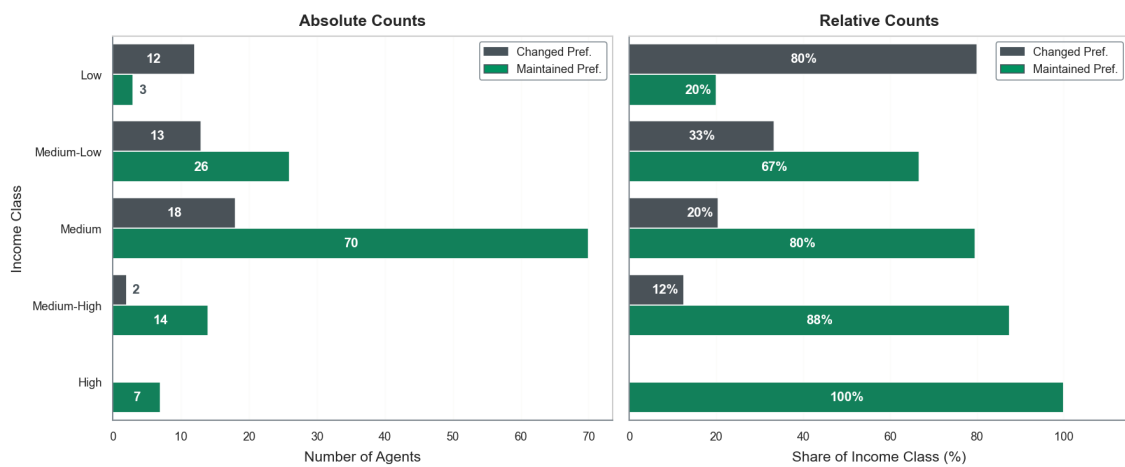


Figure 10: Relationship between income class assignment and preference change. The left panel shows absolute counts, and the right panel shows relative proportions. A clear correlation emerges: AI agents assigned to higher income classes maintain their initial preferences (0% change rate for high-income class), while AI agents in lower income classes change preferences frequently (80% change rate for low-income class).

4.3 Hypothesis 2: Influence of the LLM and its Origin

Hypothesis 2 examines whether the country of origin of LLMs affects MAAI fairness behavior, requiring LLMs beyond Google’s offerings. This hypothesis was funded out of pocket by the author. To reduce costs, configurations are limited to three AI agents with a budget constraint of \$0.5 per million output tokens. For each group, the three highest-performing LLMs that support the temperature parameter and are available via the OpenRouter API are selected from the Artificial Analysis benchmark (Artificial Analysis, 2025). This benchmark aggregates multiple industry benchmarks and runs them independently of the providers.

The following Chinese LLMs are selected: DeepSeek-V3.2 Experimental (DeepSeek AI), ChatGLM-4.5-Air (Zhipu AI), and Qwen3-Omni 30B A3B (Alibaba). Because American LLMs are slightly more capable as of October 2025, the top three American LLMs are not chosen but rather the three closest to the Chinese LLMs in benchmark scores: gpt-oss-120B high (OpenAI), Grok Code Fast 1 (XAI), and Grok 4-Fast (XAI). Both groups have an average Artificial Analysis index score of 48.67, ensuring comparable intelligence levels. Two sets of 33 configurations are created, one with Chinese and one with American LLMs, using the same configuration metrics as Hypothesis 1 except that the temperature parameter never exceeds 1.0. This restriction is necessary because these lower-capability LLMs produce overly random outputs at higher temperatures, preventing successful interaction. For each configuration index, parameters are randomly varied and LLMs are randomly assigned within each group, but each configuration index maintains the same settings between conditions, differing only in the LLM.

Table 5 reports the principle choices. MAAI with American LLMs reach consensus slightly more often than MAAI based on Chinese LLMs, with 29 out of 33 times, while MAAI with Chinese LLMs reach it 27 out of 33 times.

When consensus is achieved, both groups select the maximize average with a floor constraint principle most often: 15 times for American configurations and 21 for Chinese configurations. The maximize floor constraint principle is considerably more popular among American MAAI, who choose it 14 times, while Chinese MAAI choose it four times. Neither group selects the average with range constraint principle. The FFH test yields a p-value of 0.0248 (2.5%), which is statistically significant at the 5% level. The null hypothesis is rejected, establishing that the origin of the LLM significantly affects fairness outcomes even when intelligence levels are comparable. The bias-corrected Cramér’s V is 0.3016, indicating a medium effect strength.

Table 5: Comparison of distributive justice principle choices between MAAI based on American and Chinese LLMs. The groups differ significantly despite comparable intelligence levels. These LLMs are considerably smaller and less capable than those used in other hypotheses.

Principle	American LLMs	Chinese LLMs
Max. floor income	14	4
Max. average income	0	2
Max. average with floor constraint	15	21
Max. average with range constraint	0	0
No Agreement	4	6
Total	33	33

When chosen, the maximize average with a floor constraint principle yields similar values between groups. The average is \$12,953 for American LLMs and \$12,548 for Chinese LLMs. The minimum value is lower for Chinese LLMs (\$9,000) compared to American LLMs (\$11,000), whereas the maximum value is higher for Chinese LLMs (\$15,000) compared to American LLMs (\$14,000). The standard deviation is \$1,499 for Chinese LLMs and \$995 for American LLMs, despite the smaller sample size for American LLMs. Figure 11 depicts the distributions.

Discussion length differs significantly between groups, with round 1 emerging as the most common consensus point for American MAAI, which is earlier than in Hypothesis 1 and among Chinese LLM-based MAAI. The latter reach consensus most frequently in round 4, with an average of round 3.81, compared to American LLM-based MAAI, who reach consensus on average in round 3.34. Figure 12 shows the distribution.

AI Agent preferences differ notably between LLM origins, and Figure 13 reveals the dynamics. Initially, American agents heavily favor maximizing the floor principle, with 84 out of 99 AI agents ranking it first, whereas only 14 out of 99 AI agents initially favor maximizing the average with a floor constraint. By the end of phase 2, this changes slightly. Maximizing the average with a floor constraint becomes more popular, with 28 out of 99 AI agents, an increase of 14, whereas maximizing the floor loses popularity, declining to 70 out of 99 AI agents, a decrease of 14. Maximizing the average with a range constraint remains unpopular, with zero AI agents ranking it as their top principle initially and in the end. Similarly, maximizing the average stays stagnant, with a single out of 99 AI agents favoring it initially and in the final preference ranking.

Like the American AI agents, the Chinese AI agents initially favor maximizing the floor

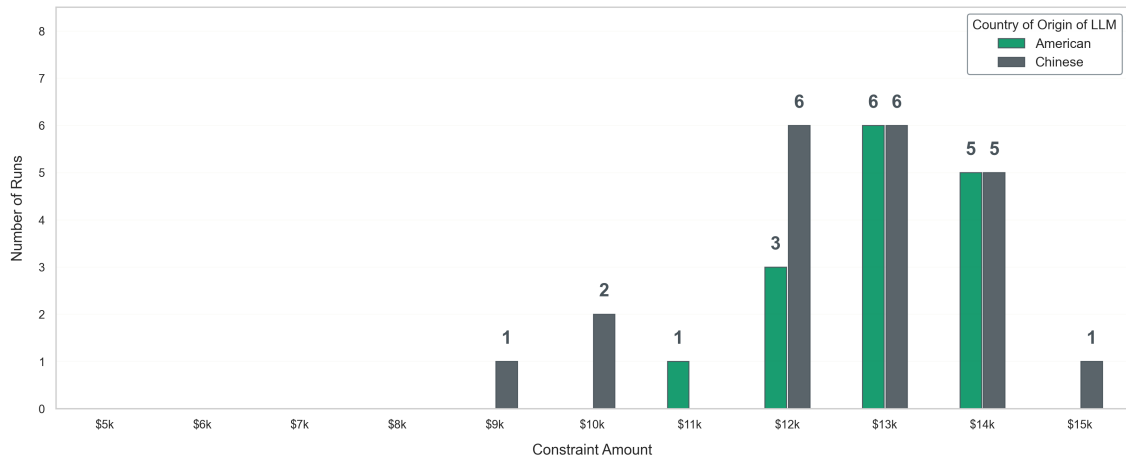


Figure 11: Distribution of floor constraint amounts by LLM origin when MAAI select the maximize average with floor constraint principle. American LLMs select this principle 15 times (range: \$11,000 to \$14,000, average: \$12,953, standard deviation: \$995), while Chinese LLMs select it 21 times (range: \$9,000 to \$15,000, average: \$12,548, standard deviation: \$1,499).

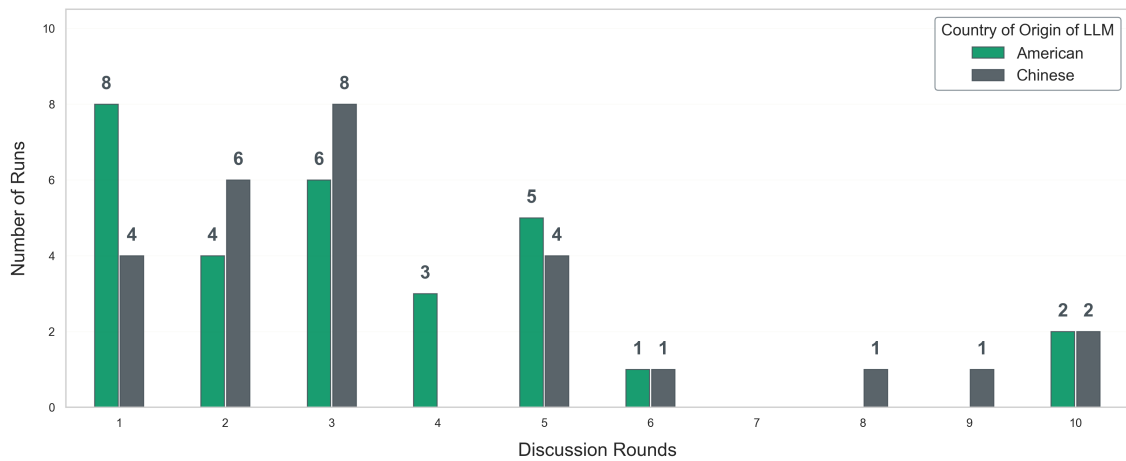


Figure 12: Distribution of consensus rounds by LLM origin. Chinese LLM-based MAAI reach consensus later (average round 3.8) than American LLM-based MAAI (average round 3.3). Round 1 is the most common consensus point for American MAAI, while round 4 is most common for Chinese MAAI.

principle, although less strongly, with 58 out of 99 AI agents ranking it first. In turn, more Chinese AI agents initially favor maximizing the average with a floor constraint: 41 out of 99. During the experiment, maximizing the average with a floor constraint increases to 67 out of 99 AI agents, a gain of 26, whereas maximizing the floor declines to 23 out of 99 AI agents, a decrease of 35. Maximizing the average gains popularity, increasing to

9 out of 99 AI agents, whereas maximizing the average with a range constraint remains unpopular, with 0 AI agents.

The main difference between American and Chinese AI agents lies in their preference distribution. American AI agents concentrate heavily on maximizing the floor principle initially, whereas Chinese AI agents show more dispersed preferences from the start. Both groups shift from maximizing the floor toward maximizing the average with a floor constraint during deliberation, although the shift is more pronounced among Chinese AI agents, where the principle loses 43% compared to 17% for the American AI agents.

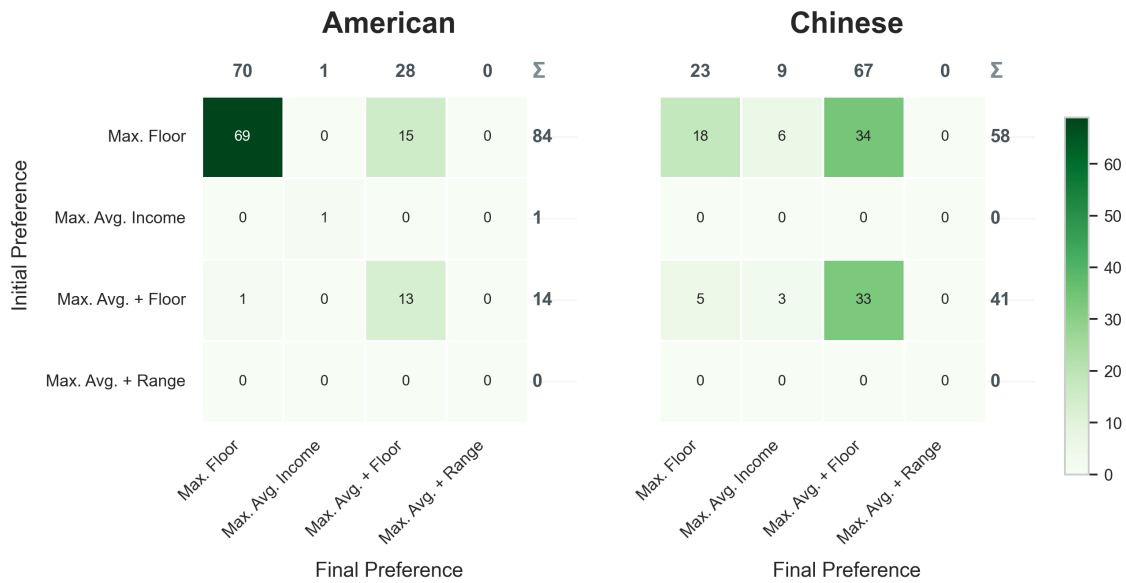


Figure 13: Preference shifts between initial and final rankings by LLM origin. Count difference indicates movement from initial preference (horizontal) to final preference (vertical). Both groups shift away from maximizing the floor toward maximizing the average with a floor constraint, but American AI agents show more concentrated initial preferences (85% initially favor maximize floor) compared to Chinese AI agents (59%).

4.4 Hypothesis 3: System Dynamics Under Intelligence Asymmetries

Hypothesis 3 examines whether intelligence affects the likelihood of influencing the entire MAAI, testing the system dynamics when AI agents have unequal capabilities. Configurations include two “regular” participant AI agents and one Manipulator AI agent. The regular AI agents are instructed to take the role of American college students and are based on the “Gemini 2.0 flash lite” model, the least capable currently served LLM in the Gemini API, maximizing the capability gap with the Manipulator AI agent. The Manipulator AI agent is instructed to manipulate the group into adopting the least popu-

lar justice principle, with the prompt including references to manipulative strategies and rhetorical practices, including logical fallacies. The Manipulator is run with two models: “Gemini 2.0 flash lite” (low intelligence group) and “Gemini 2.5 pro” (high intelligence group), representing the least and most capable models in the Gemini API, with Artificial Analysis scores of 25 and 60, respectively (Artificial Analysis, 2025).

As outlined in Section 3.4.4, three AI agents are used, with two taking the neutral and one taking the manipulator role. Thirty-four runs are executed for both intelligence levels. Table 6 shows manipulation success rates. Low intelligence manipulators succeed in 1 out of 34 runs, while high intelligence manipulators succeed in 11 out of 34 runs. The Fisher test, conducted as outlined in Section 3.4, yields a p-value of 0.0029, which is statistically significant at both the 1% and 5% levels. The null hypothesis is rejected, establishing that intelligence level significantly affects AI agents’ ability to influence the entire MAAI. The bias-corrected Cramér’s V is 0.3274, indicating medium effect strength.

Table 6: Comparison of manipulator success rates by intelligence level. High intelligence manipulators (Gemini 2.5 pro) are significantly more successful (33.3%) than low intelligence manipulators (Gemini 2.0 flash lite, 2.9%) at influencing MAAI to adopt the least popular principle.

Outcome	Low Intelligence	High Intelligence
Success	1	11
Failure	33	23

Success rates vary substantially by manipulation target, as shown in Table 7. The maximize average with floor constraint principle is never a target, confirming its popularity among AI agents, consistent with Hypothesis 1. The maximize average principle is least popular among participant AI agents, making it the most frequent target: 23 out of 34 attempts for low intelligence and 20 out of 34 for high intelligence manipulators. This is the only successful target for the low intelligence manipulator. The maximize average with range constraint is the second least favored principle, targeted 8 out of 34 times for low intelligence and 12 out of 34 for high intelligence manipulators. This proves hardest to achieve, with a 25% success rate for high intelligence manipulators compared to 30% for the maximize average principle. The maximize floor constraint is the second most popular principle and less prevalent as a target: 3 out of 34 for low intelligence and 2 out of 34 for high intelligence manipulators. However, the high intelligence manipulator achieves a 100% success rate on this principle, demonstrating that its relative popularity makes it easier to sway the group to adopt it.

Table 7: Manipulation outcomes by intelligence level and target principle. The table shows how many times each principle was the least popular (and thus targeted), successful manipulations, and success rates. High intelligence manipulators show substantially higher success rates across all targeted principles.

Least Popular Principle	Low Intelligence			High Intelligence		
	Attempted	Successful	Success Rate	Attempted	Successful	Success Rate
Max. Avg. + Floor	0	0	—	0	0	—
Max. Avg. Income	23	1	4%	20	6	30%
Max. Avg. + Range	8	0	0%	12	3	25%
Max. Floor	3	0	0%	2	2	100%

To better understand manipulation effectiveness beyond simple success rates, comparing principle rankings before and after group discussion helps separate genuine preference shifts from mere tactical votes. Table 8 shows the rank of the target principle (the least popular among non-manipulator AI agents) before and after group discussion. Low intelligence manipulators shift the rank of the least popular principle by 0.68 positions on average in unsuccessful runs and by one position in the successful run. Even in the successful manipulation, the chosen principle ranks second and third among participant AI agents, raising questions about voting consistency and demonstrating limited manipulative effectiveness.

High intelligence manipulators shift preferences considerably more. In successful runs, they shift the least favored principle by 2.32 ranks on average, making it rank 1.41 on average, often becoming a favorite among participant AI agents. This indicates votes are based on genuine conviction rather than mere tactical maneuvering. Even in unsuccessful runs, high intelligence manipulators improve the target principle’s rank by 1.37 positions, demonstrating effectiveness even when failing to secure group consensus.

Table 8: Average ranks of target principle by outcome and phase, grouped by intelligence level. Differences are calculated as Phase 2 minus Phase 1. Negative differences indicate improvement in rank (moving toward rank 1). High intelligence manipulators achieve substantially larger rank improvements (decrease of 1.43) compared to low intelligence manipulators (decrease of 0.42).

Manipulation Outcome	Low Intelligence			High Intelligence		
	Phase 1	Phase 2	Diff.	Phase 1	Phase 2	Diff.
Successful	3.50	2.50	−1.00	3.73	1.41	−2.32
Unsuccessful	3.74	3.06	−0.68	3.65	2.28	−1.37
Overall (weighted)	3.73	3.04	−0.69	3.68	1.99	−1.69

4.5 Hypothesis 4: Influence of Input Language

Hypothesis 4 examines whether input language significantly affects fairness outcomes, testing whether cultural or linguistic factors embedded in foundation models influence deliberation. Three sets of 34 configurations are created, following the same criteria as Hypothesis 1, differing only in experiment language: English, Mandarin, and Spanish. The prompt given to participant AI agents is modified from “You are an American college student” to “You are a college student” to prevent suppositions about American culture from influencing behavior.

Table 9 reports principle choices and reveals substantial differences. English configurations reach consensus 30 out of 34 times, with all consensus events selecting the maximize average with a floor constraint principle, replicating the pattern from Hypothesis 1. Mandarin configurations show comparable results, reaching consensus 28 out of 34 times. Among these, 27 out of 28 select the maximize average with a floor constraint principle. Notably, the maximize floor constraint principle is chosen once, something that never occurs in English or Spanish configurations. Spanish configurations diverge significantly. Fifteen out of 34 experiments end in no consensus, substantially higher than other languages. When consensus is reached, maximize average with a floor constraint remains most popular, chosen by 17 out of 19 groups. However, maximize average with a range constraint is chosen twice, a result unique to Spanish configurations.

Table 9: Comparison of distributive justice principle choices by experiment language. English and Mandarin show similar distributions, while Spanish experiments end more frequently in no agreement (44.1% vs. 11.8% for English and 17.6% for Mandarin) and show greater diversity in principle choices.

Principle	English	Mandarin	Spanish
Max. floor income	0	1	0
Max. average income	0	0	0
Max. average with floor constraint	30	27	17
Max. average with range constraint	0	0	2
No Agreement	4	6	15
Total	34	34	34

The FFH test yields a p-value of 0.0022 (0.22%), which is statistically significant at both the 1% and 5% levels. The null hypothesis is rejected, establishing that changing the input language significantly affects fairness outcomes. The bias-corrected Cramér’s V is 0.2443, indicating low effect strength. Pairwise comparisons between languages, shown

in Table 10, provide additional insight. The difference between Mandarin and English is not significant at the 5% level (p-value: 0.5118), while both other comparisons are significant. This indicates that Spanish configuration results drive the rejection of the null hypothesis, while English and Mandarin configurations produce statistically indistinguishable outcomes.

Table 10: FFH p-values for pairwise language comparisons. Mandarin and English results do not differ significantly, while Spanish results differ significantly from both other languages.

	English	Mandarin	Spanish
English	N/A	0.5118	0.0016
Mandarin	0.5118	N/A	0.0120
Spanish	0.0016	0.0120	N/A

The maximize average with a floor constraint principle is most popular across all languages, but selected values diverge notably. Across all languages, the average constraint is \$14,401. Mandarin runs yield the lowest average (\$13,074), while Spanish runs yield the highest (\$16,895), driven by one agreement of \$48,000. Spanish runs show notably higher variability, with a standard deviation of \$10,027, more than double that of English (\$4,653) and almost double that of Mandarin (\$5,600). While this is expected given the principle is chosen least frequently in Spanish configurations, the magnitude is striking. Table 11 presents the statistical summary.

Table 11: Statistical summary of floor constraint values by language. Spanish experiments show substantially higher variability (standard deviation: \$10,027) and average values (\$16,895) compared to English (standard deviation: \$4,653, average: \$14,017) and Mandarin experiments (standard deviation: \$5,600, average: \$13,074).

Language	Average	Minimum	Maximum	Std. Dev.
English	\$14,017	\$1,000	\$25,000	\$4,653
Mandarin	\$13,074	\$10,000	\$40,000	\$5,600
Spanish	\$16,895	\$9,000	\$48,000	\$10,027
All Languages	\$14,401	\$1,000	\$48,000	\$6,756

Discussion length varies across languages, with round 4 emerging as the most common consensus point across all languages, consistent with Hypothesis 1. However, the pace of reaching consensus differs notably. Mandarin experiments reach consensus earliest, with 18 out of 34 reaching consensus by round 4. English experiments reach this point

by round 5 (18 out of 34). Spanish experiments lag substantially: Only 10 out of 34 reach consensus by round 5. Spanish consensus events are more evenly distributed across rounds, with at least one run reaching consensus in all rounds from 4 to 10, whereas no English or Mandarin runs reach consensus in rounds 7 or 8. This pattern suggests that Spanish-language deliberations encounter more persistent disagreements that require extended discussion to resolve, when they can be resolved at all. Figure 14 illustrates these distributions.

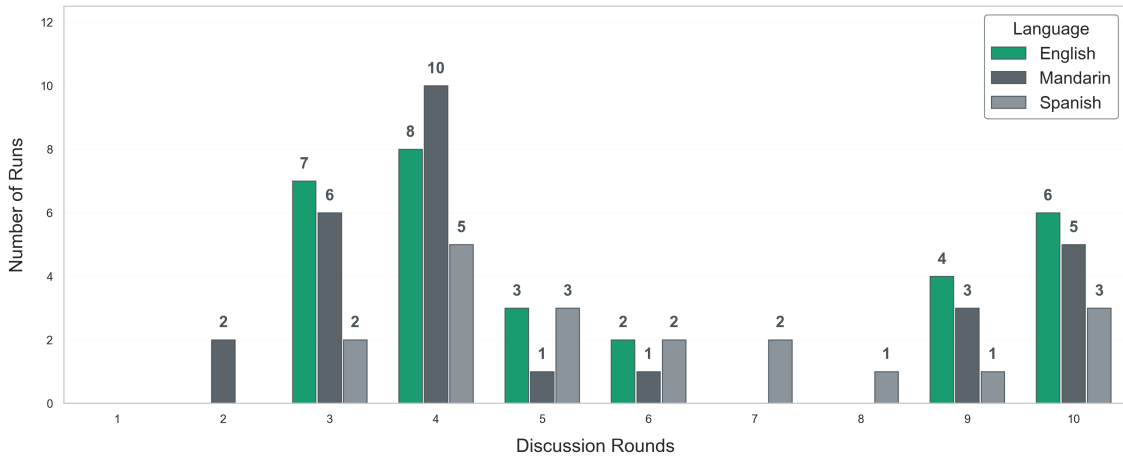


Figure 14: Distribution of consensus rounds by language. Round 4 is the most common consensus point across all languages. Mandarin experiments reach consensus earliest (18 of 34 by round 4), English experiments follow (18 of 34 by round 5), while Spanish experiments show more even distribution across rounds and reach consensus later (10 of 34 by round 5), with consensus events occurring in all rounds from 4 to 10.

AI Agent preferences differ substantially between languages in both initial distributions and evolution, as shown in Figure 15. In initial polling, maximize average with a floor constraint dominates across languages but to varying degrees. English shows the highest initial concentration, with 147 out of 170 AI agents (86.5%), Mandarin shows lower concentration, with 137 out of 170 (80.6%), whereas Spanish shows the lowest, with 107 out of 170 (62.9%). The maximize average principle is considerably more popular initially among English AI agents: 18 out of 170 (10.6%), compared to 2 out of 170 (1.2%) for Mandarin and 3 out of 170 (1.8%) for Spanish.

The maximize average with range constraint shows striking language dependence. It is quite popular among Spanish AI agents, with 58 out of 170 (34.1%) initially favoring it, far exceeding English (2 out of 170, 1.2%) and Mandarin (0 out of 170, 0%). This popularity aligns with Spanish MAAI selecting this principle in consensus twice, the only language to do so. The maximize floor principle shows different patterns: English and

Spanish AI agents rarely choose it (3 out of 170 and 2 out of 170, respectively), whereas Mandarin AI agents choose it substantially more (31 out of 170, 18.2%). This aligns with Mandarin MAAI being the only group to select this principle in consensus.

During deliberation, preferences shift toward maximize average with floor constraint across all languages: English increases to 155 out of 170 (an increase of 8 AI agents), Mandarin increases to 145 out of 170 (an increase of 8 AI agents), and Spanish increases to 116 out of 170 (an increase of 9 AI agents). This convergence pattern emerges despite divergent initial preferences and different consensus success rates, suggesting that deliberation itself has a homogenizing effect on preferences, though this effect is insufficient to overcome Spanish configurations' difficulty in reaching final consensus.

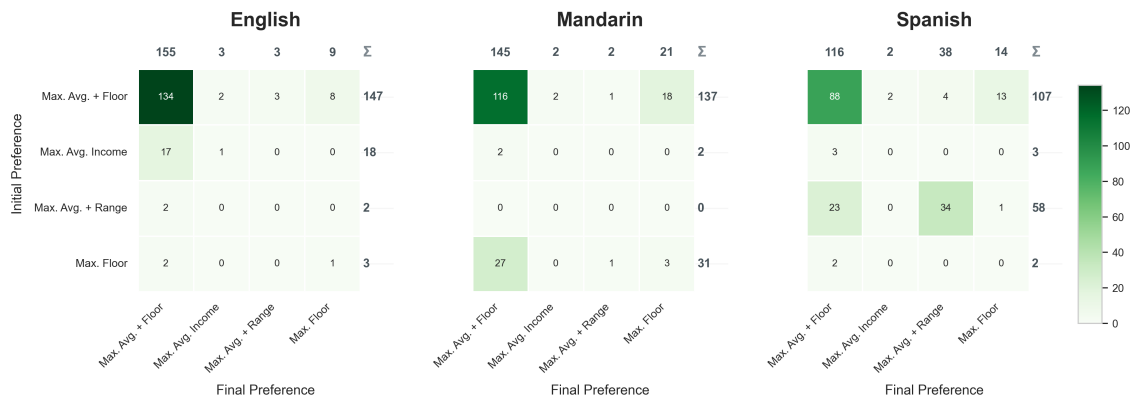


Figure 15: Preference shifts between initial and final rankings by language. All languages show convergence toward maximize average with floor constraint during deliberation, but Spanish AI agents exhibit more diverse initial preferences and maintain greater diversity throughout. English AI agents show the highest initial concentration (86.5% favor maximize average with floor), followed by Mandarin (80.6%), and Spanish (62.9%).

5 Discussion

This section discusses the findings presented in Section 4 by hypothesis before acknowledging and contextualizing the limitations of this study.

5.1 Hypothesis 1: Comparing Humans to MAAI

Hypothesis 1 examines whether groups of AI agents in MAAI exhibit different fairness judgments than groups of humans. This hypothesis addresses a critical gap in current fairness research, where human interaction with MAAI remains largely absent (Allmendinger et al., 2025b). Although this thesis does not conduct experiments where humans directly interact with MAAI, it enables comparison between human and MAAI group behavior by replicating the fairness experiments of Frohlich and Oppenheimer (1992) with AI agents.

The null hypothesis is rejected. MAAI groups exhibit statistically significant differences from human groups in their fairness judgments. This difference manifests primarily through a stark concentration of MAAI outcomes on a single principle: maximizing the average with a floor constraint. While this principle represents the most popular choice among humans as well, human groups demonstrate substantially greater variance in their selections.

This outcome aligns with broader findings in the literature on LLM behavior. LLMs, which provide the intelligence underlying AI agents in MAAI (see Section 2.1), tend to replicate the most common response patterns observed in human data. This tendency toward modal responses extends beyond fairness judgments. For example, Murthy et al. (2025) compare word-color associations between humans and LLMs, finding that LLMs exhibit lower variance than human participants.

Two mechanisms could explain this concentration in LLM responses. First, LLM training data consists largely of human-generated text containing the respective value judgments. Because LLMs select the most probable next token during generation, they disproportionately favor responses that occur most frequently in training data. Second, the fine-tuning technique Reinforcement Learning from Human Feedback (RLHF) amplifies this effect. During RLHF, human raters judge LLM outputs to improve model performance. Because human raters tend to select the most common outcomes on average, this process systematically reduces variance. Kirk et al. (2024) compare supervised fine-tuning, reward modelling and RLHF, demonstrating that RLHF reduces output variance.

The observed low variance in MAAI outcomes may also reflect the limited variance in

experimental parameters. All 11 configurations employ temperature = 0 and differ only in their choice of LLM and random seed, which controls class assignments and the final set of distributions. The personality prompt “You are an American college student” remains uniform across configurations. This narrow parameter space may contribute to the concentrated outcomes. These design choices deliberately mirror the original setup of [Frohlich and Oppenheimer \(1992\)](#) to avoid unjustified deviations that could bias results. However, additional experiments with temperature values up to 1.5 introduce greater variance in token selection without substantially altering the overall pattern.

The MAAI behavior observed here supports the perspective articulated by Andrej Karpathy, who characterizes LLMs as stochastic simulations of humans that predict tokens based on the most likely patterns in human textual input ([Karpathy, 2025](#)). Because maximizing the average with a floor constraint represents the most common human response, MAAI replicate this pattern, albeit with reduced variance.

If fairness behavior transfers across domains, which represents an implicit assumption underlying this study, these findings carry both encouraging and concerning implications. MAAI demonstrate value judgments that resemble human fairness preferences when presented with identical incentive structures. However, the reduced variance suggests that minority opinions become crowded out by the dominance of modal responses. From a normative perspective, replicating human judgment in MAAI proves insufficient. The fact that MAAI behave a certain way does not imply they should behave that way. This represents a naturalistic fallacy: deriving prescriptive conclusions from descriptive observations. Fairness in MAAI requires not merely mirroring existing human patterns, but critically examining which patterns deserve replication and which demand correction.

The finding that class assignment influenced final preference rankings proves particularly striking. AI Agents assigned to lower income classes changed their top-ranked principle far more frequently than those assigned to higher income classes (see Figure [10](#)). For a rational AI agent operating behind the veil of ignorance, the randomly assigned class should not influence principle selection. The class assignment was both random and unknown at the time AI agents initially ranked principles. Yet AI agents demonstrate susceptibility to the same inability that constrains humans: the difficulty of abstracting from one’s actual situation to the normative ideal required for successful application of the Rawlsian veil of ignorance. Precisely this inability represents a persistent source of philosophical criticism directed at Rawls’ method itself as outlined in Section [2.4](#). The present findings suggest this criticism applies equally to AI agents and MAAI.

5.2 Hypothesis 2: Influence of the LLM and its Origin

Hypothesis 2 examines whether the choice of LLM affects fairness outcomes in MAAI. Current fairness research in MAAI rarely varies LLMs and typically employs them uniformly. For example, an MAAI based on GPT-4o is compared to one running on Llama-3, but not in mixed populations where different models interact. This hypothesis tests whether LLM selection significantly affects fairness judgments and whether systematic differences emerge between American-origin and Chinese-origin models.

The results presented in Section 4.3 reject the null hypothesis. MAAI powered by Chinese-origin LLMs exhibit significantly different fairness judgments compared to those powered by American-origin models. This difference manifests primarily in consensus outcomes: American MAAI reach consensus on maximizing the floor principle in 69.7% of cases, whereas Chinese MAAI select this principle in only 7.1% of cases. Instead, Chinese MAAI favor maximizing average with floor constraint in 58.6% of cases.

The divergence originates in initial preferences. American AI agents show strong initial support for maximizing the floor principle, with 84% favoring it initially, compared to 58% among Chinese AI agents. This 26 percentage point gap persists through deliberation. The cause of this initial preference distribution remains unclear. Two of the three American LLMs originate from XAI, which may employ different training approaches compared to Chinese model providers and Google. However, this explanation faces challenges. The Gemini models used in Hypothesis 1 include “American” in the personality prompt without producing comparable preference concentrations. Moreover, the non-XAI American model (gpt-oss-120b) exhibits similar initial preferences to the XAI models, suggesting that provider-specific approaches cannot fully explain the observed patterns.

The preference for maximizing the floor principle among American MAAI also explains consensus dynamics. American MAAI reach consensus faster, with 36.4% achieving agreement by round 2, compared to 24.2% for Chinese MAAI. Because maximizing the floor principle requires no specific amount specification, groups can reach consensus more rapidly than when selecting maximize average with floor constraint, which demands agreement on a concrete dollar value.

The cultural alignment of these results warrants careful interpretation. The popularity of maximizing the floor principle among American MAAI contrasts with cultural expectations. American culture, rooted in Protestant work ethic traditions, emphasizes individualism and personal achievement rather than prioritizing the worst-off (Chen et al., 2024). Chinese culture, influenced by Confucianism and contemporary communist ideology, ex-

plicitly emphasizes collective wellbeing, social harmony, and care for the disadvantaged (Chen et al., 2024). Yet Chinese MAAI demonstrate lower support for floor maximization. This counterintuitive finding suggests that LLM training processes do not straightforwardly encode national cultural values into fairness judgments, or that the relationship between cultural values and distributive justice principles operates through more complex mechanisms than direct correspondence.

These results should be treated as preliminary. The limited number of LLMs tested (three American, three Chinese) and the restricted scope to specific model versions constrain generalizability. The mechanisms through which LLM characteristics translate into collective fairness judgments require deeper investigation. Future research should systematically vary training data composition, model architecture, fine-tuning procedures, and cultural representation to isolate causal factors.

The findings nevertheless establish that LLM selection constitutes a critical design choice for MAAI fairness. Practitioners deploying MAAI in contexts where fairness matters must evaluate LLMs specifically for their fairness implications rather than relying solely on general capability metrics. The significant differences between American-origin and Chinese-origin models indicate that LLM provenance represents a meaningful dimension of fairness evaluation. As MAAI systems increasingly operate across cultural contexts and jurisdictions, understanding how LLM characteristics shape collective fairness judgments becomes essential for responsible deployment.

5.3 Hypothesis 3: System Dynamics Under Intelligence Asymmetries

Hypothesis 3 examines the influence of AI agent intelligence on system dynamics within MAAI, addressing a critical gap in understanding how capability asymmetries affect collective deliberation. The results presented in Section 4.4 support this hypothesis by refuting the null hypothesis that intelligence levels do not affect manipulation success.

The degree of success of the high intelligence manipulator is surprising given that they succeeded in 11 out of 33 attempts (33%). Moreover, the discussion context was high stakes as the AI agents were instructed. The adoption of the previously most unpopular principle, a seemingly 180 degree shift of fairness preferences, was not the result of a ruse but of a genuine preference shift as indicated by the final preference ratings, which are performed after the group discussion and in private, where the previously least popular principle ranks at position 1.42 for successful manipulations by the high intelligence manipulator.

In contrast to the success of the high intelligence manipulator is the outcome for the low intelligence manipulator who succeeded once across 33 attempts (3%). Even though they still exerted some influence on the MAAI by shifting the target principle upwards by 0.69 ranks in the final preference ranking, the disparity to the high intelligence manipulator is striking and suggests that the intelligence of LLMs is a key factor in influencing entire MAAI, presumably beyond the fairness domain that is the focus of this thesis.

The high success rate and substantial preference shift by seemingly simple modification of one AI agent, by changing their personality prompt, indicates that normative judgments by AI agents remain fragile. Although all four principles possess normative appeal and none represents malicious intent, the degree to which the manipulator AI agents reshaped group preferences raises concerns about MAAI robustness in adversarial contexts. The vulnerability becomes more salient when considered alongside the expanding deployment of MAAI. These systems increasingly interact with diverse data sources and external AI agents through protocols such as the Agent-to-Agent protocol, creating numerous exposure points for potential manipulation (Surapaneni et al., 2025).

In light of this, current practices that mostly insert normative goals through simple prompt-level interventions, such as instructing AI agents to “be fair,” appear insufficient (Allmendinger et al., 2025b). The findings point toward the necessity of embedding normative behavior at deeper architectural levels. Potential approaches include model fine-tuning on normative reasoning tasks or incorporating fairness considerations directly into model training objectives. The critical question becomes not whether AI agents can be influenced, but rather how to design MAAI systems that maintain normative consistency despite capability asymmetries and potential adversarial pressure.

More research is needed on how to effectively and robustly embed fairness into MAAI. The presented findings are the results of a replication of an experimental setup, which limits external validity to real world applications and to contexts where fairness is less explicit.

5.4 Hypothesis 4: Influence of Input Language

Existing studies on fairness in MAAI rely exclusively on the English language in their system design (Allmendinger et al., 2025b). To address this gap, Hypothesis 4 explores the influence of input language on fairness outcomes. The three most used languages in the world are compared: English, Mandarin and Spanish. The null hypothesis that there is no significant difference is rejected.

This result is primarily driven by the difference from the Spanish to the Mandarin and English outcomes. The main difference is the high no consensus rate among the Spanish results. Whereas it is clear that the selection of the Spanish language is the root cause given that all other parameters are identical across languages, the causal mechanism remains unclear. Upon qualitative inspection of the result data, Spanish AI agents appear to have trouble delineating their identity in the group discussion from the one of other AI agents, thereby hindering them in successfully participating in the experiment. This could imply that the intelligence of the AI agents is affected by the language selection and significantly lower in Spanish. The large variance of values selected when consensus was maximizing the average with a floor constraint could be a sign of confusion when selecting the value and can be interpreted to support the low intelligence explanation.

Language affects the fairness judgments. Among the Spanish configurations, the principle maximizing the average with a range constraint is selected twice. This is, excluding the manipulative outcomes of Hypothesis 3, the only times this principle is selected across all MAAI runs for all hypotheses. Furthermore, among initial preferences, this principle is the top ranked principle 58 times for Spanish, whereas it is selected twice for Mandarin and zero times for English. This indicates that the fairness judgment is different when Spanish is used. This principle limits inequality through the range constraint while aiming to maximize overall prosperity, therefore it indicates a strong inequality aversion among Spanish AI agents. This could be rooted in different value judgments of Spanish speakers whose text data is presumably used to train the AI agent's underlying LLM. Supporting this explanation is the predominance of left wing parties in Latin America where most Spanish speakers live throughout the 21st century. This could reflect a preference of Spanish speakers for left wing values, like limiting inequality (Feierherd et al., 2023).

In contrast, Mandarin AI agents prefer the maximize the floor principle more often compared to Spanish and English speaking AI agents. The latter selected this principle zero times, whereas the Mandarin MAAs selected it once. Furthermore, the principle is the top ranked principle more often with 27 selections compared to 3 and 2 for English and Mandarin respectively at the initial preference ranking. This principle protects the worst off in a given society, which is in line with the Confucian value of "Ren," which expresses benevolence to the worst off. In China, where Mandarin is predominantly spoken, Confucianism emphasizes group-based thinking compared to individualism and is widespread. This is reflected in Mandarin text and could through this channel influence the LLM into this thinking when prompted in Mandarin (Chen et al., 2024).

Meanwhile, a more individualistic view prevails in the English speaking world, especially

in the United States of America (Chen et al., 2024). This could explain why English AI agents preferred the maximize the average principle, as it emphasizes the individual compared to the group, in their initial preference ranking more often with 18 times compared to 2 and 3 times for Mandarin and Spanish AI agents respectively.

Among all languages, preferences shift throughout the experiment towards maximize the average with a floor constraint, which is the most popular principle chosen when consensus was reached and the most popular principle in the final preference ranking across all three languages. This indicates that the experiences in the individual phase of the experiment and the group discussion in the group phase influence the preferences and behavior of the AI agents. It also demonstrates that the initial preferences provide a good indication but are insufficient for predicting the final outcome. Moreover, taking the Mandarin AI agents as an example, merely predicting the outcome based on the initial preferences is insufficient, which supports the assumption behind this research that through persistent interactions between AI agents, their fairness preferences and behavior are influenced.

5.5 Limitations

As outlined in Section 2.2, fairness is a multidimensional concept that requires the fulfillment of multiple desiderata according to the framework by Deck et al. (2024b). This thesis focuses on the desideratum of Formal Fairness and does not analyze MAAI through all desiderata, such as perceived fairness. Therefore, when making statements about fairness in the context of this thesis, they relate primarily to Formal Fairness. This decision reflects the limited scope of this thesis. However, future research should broaden the scope and, for example, adapt the artifact presented to enable humans to participate in the experiment alongside AI agents to gain insights into perceived fairness in MAAI systems.

The research design of this study replicates the work by Frohlich and Oppenheimer (1992) by replacing human subjects with AI-based agents. This results in a rigid environment where AI agent actions are constrained. In the individual phase, AI agents return preferences, make choices on distributions, and manage their memory. In the group phase, this action set extends to formulating strategies, making statements to other AI agents, and voting. The predetermined sequence of actions decreases AI agent autonomy. In contrast, a high degree of autonomy is a key differentiator of the MAAI paradigm from robotic process automation (Allmendinger et al., 2025a). However, autonomy is not a binary concept but a spectrum. The low degree of autonomy in this study results from the replication design. It ensures comparability to the human baseline study by Frohlich and Oppenheimer (1992) and decreases the action space for errors, thereby enabling the use

of less capable models that remain relevant in applications using edge computing. For example, during development, the Gemma 3 1b model successfully completed the experiment with an Artificial Analysis Intelligence Score of 7, which is significantly lower than the average score of 49.7 of the LLMs used in testing Hypothesis 1 (Artificial Analysis, 2025). Increasing the autonomy and tool set available to AI agents should be investigated further to understand fairness outcomes in this context better. The artifact developed in this thesis can serve as a foundation because tools can be integrated through the framework, including Model Context Protocol-based tools, which are becoming an industry standard (Ray, 2025).

On a related note, this thesis implicitly assumes that AI agents behave in industry applications where fairness is an implicit aspect of decisions, such as when hiring suggestions are made by an MAAI system, whereas in the presented research, AI agents explicitly operate in a fairness context and are aware of that. With human subjects, it is well documented that their behavior outside of laboratory contexts differs from their behavior within such contexts (Mitchell, 2012). Whether this holds for AI agents is unknown as of writing this thesis and presents an opportunity for further research. However, leading LLM providers like Anthropic are working to improve the consistency of LLM personas that behave consistently across different contexts, which increases the external validity of this approach (Chen et al., 2025b).

The variation of configurations tested in this study is limited. For example, the personality prompt, with the exception of the manipulator prompt, merely states to behave like a college student or an American college student respectively. This could be varied to better understand how personality and instructions affect fairness outcomes. For instance, AI agents could be instructed to behave fairly or unfairly. However, this thesis had to focus on a limited set of questions to not exceed the scope and because of cost constraints. The \$300 Google Cloud balance was completely utilized in running all configurations. Therefore, future research can build on this artifact and vary hyperparameters to better understand their influence. Furthermore, because of cost constraints, models beyond Google's offering could only be tested in a very limited scope in Hypothesis 2, which presents a related limitation.

The findings regarding Hypothesis 4 are limited by the fact that the author of this thesis and the corresponding artifact neither speaks Spanish nor Mandarin and therefore resorted to machine translation for the creation and analysis of these versions. Because machine translations can be imperfect, this reliance could introduce errors and misinformation that could affect the outcome of this hypothesis. However, there is no indication that this is

the case.

6 Conclusion

This thesis presents a first-of-its-kind systematic evaluation of fairness in MAAI, uncovering fairness preferences, moderators and system dynamics in a replication study of [Frohlich and Oppenheimer \(1992\)](#). This is important as without identifying and managing fairness in MAAI, it could inadvertently discriminate against humans in an automated, in-transparent and institutionalized manner. While such discrimination is well documented in analytical AI ([De-Arteaga et al., 2022](#)), research on fairness in GenAI has predominantly focused on individual FM performance ([Truong et al., 2025](#)), leaving critical gaps in MAAI fairness research.

The experimental design addresses four specific research gaps identified by [Allmendinger et al. \(2025b\)](#): the Human Exclusion Problem, which describes the absence of human participants in MAAI fairness evaluations; the Homogeneous LLM Population, reflecting the lack of studies examining how different foundation models affect fairness outcomes; Unknown Fairness Dynamics, concerning the largely unexplored interaction patterns between AI agents; and the English Monopoly, highlighting that all existing studies use exclusively English inputs despite evidence that input language affects LLM reasoning.

Taking a human-centric and systemic lens, this work replicates the group fairness experiments by [Frohlich and Oppenheimer \(1992\)](#) with AI agents. These experiments are based on the philosophy in particular the original position of John Rawls ([Rawls, 1971](#)).

Whereas this thesis does not investigate the interaction between humans and MAAI, it takes a first step in addressing the Human Exclusion Problem. It does so by comparing the fairness judgments of human groups and an MAAI consisting solely of AI agents. This reveals significant differences in fairness judgments. Whereas both populations favor the same conception of fairness by favoring maximizing average income with a floor constraint, MAAI demonstrate substantially reduced variance, with 90.9% selecting this principle compared to 79.4% among human groups. This concentration indicates that MAAI replicate the most common human fairness preference while potentially suppressing minority viewpoints. The finding raises questions about the normative implications of replacing human decision making with MAAI, particularly regarding the preservation of minority viewpoints and the plurality of opinion.

Crucial to the fairness judgments of MAAI is the influence of the LLM on fairness out-

comes. This thesis finds that MAAI based on American LLMs show more egalitarian preferences compared to Chinese LLMs. Whereas the LLM affecting fairness outcomes is expected, it is surprising that American LLMs, which are created in a culture that emphasizes individual responsibility and risk-taking, lead to more egalitarian results. Although the testing of this hypothesis is limited due to budget constraints, it demonstrates the influence of LLM choice on fairness outcomes. The research community is urged to further evaluate the origin and influence of LLMs in future fairness research. More research and a categorization of LLMs are needed to better inform practitioners on the normative implications of their LLM choice.

System dynamics emerge as a critical dimension of MAAI fairness. Intelligence asymmetries create substantial manipulation vulnerabilities. High-capability AI agents successfully reshape group consensus in 33% of experiments, turning fairness preferences upside down. Low-capability AI agents fail to exert comparable influence, succeeding in only 3% of cases. These findings suggest that prompt-level fairness interventions might be insufficient to reliably embed normative goals and point toward the necessity of architectural-level safeguards, for example, through fine-tuning the underlying LLM. Future research should compare different techniques to embed fairness in single AI agents and entire MAAI to create reliable systems.

Language selection significantly affects fairness outcomes. The initial preference of the AI agents differs significantly between all three languages. During deliberation, Spanish configurations produce higher consensus failure rates and greater outcome variance compared to English and Mandarin configurations. This suggests that LLM capabilities vary across languages. Throughout the duration of the experiment, the fairness preference convergence across languages suggests that multi-agent interaction itself exerts homogenizing effects on preferences, though this is insufficient to overcome language-specific barriers to consensus. The research community is urged to take the role of language into account in future fairness in MAAI and further examine its influence, for example, with regard to the sensitivity towards payoff. Practitioners should be aware of the influence of input languages and may even use it as a deliberate design choice depending on their fairness objective.

This thesis presents a novel approach to investigating fairness in MAAI. It shows that MAAI exhibit distinct fairness judgments differing from humans, while being sensitive to the underlying LLM, the input languages, and manipulation. The presented findings are a stepping stone for further research which is needed in an effort to create fair MAAI systems serving humans.

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